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An alternative aerodynamic mitigation measure for improving bridge flutter and vortex induced vibration (VIV) stability: Sealed traffic barrier

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A R T I C L E I N F O

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A B S T R A C T

This study proposes an alternative aerodynamic mitigation measure, i.e., sealed traffic barrier, for bridges and verifies its efficacy in improving the flutter and vortex induced vibration (VIV) stability of bridges through wind tunnel tests for three distinctive real-world bridge projects. The results show that: (1) The measure of the sealed traffic barrier with regular intervals along the bridge deck has a significant influence on the flutter and VIV performance of the bridge. Thus, a reasonable design of the traffic barrier can simultaneously improve the bridge's flutter and VIV stability; (2) The sealed traffic barrier with vertical intervals along the bridge can effectively suppress the vertical VIVs of the bridge, while the sealed traffic barrier without vertical intervals, i.e., configured continuously along the bridge, has a better vibration suppression effect on the torsional VIV than that on the vertical VIV; (3) This proposed measure installed on both sides of the bridge deck imposes favorable suppression effect for the flutter vibration, similar to the upward central stabilizer. In case with the sealing form fixed, but a reduced height, for the barrier, the flutter stability at different wind attack angles is slightly decreased; and (4) In certain scenarios, increasing the ventilation rate of the traffic barrier results in an increase for the flutter critical wind speed at positive wind attack angles, but a decrease for the flutter critical wind speed at negative wind attack angles. This aerodynamic mitigation measure has been successfully applied to more than ten other bridge projects based on the wind tunnel tests by the leading authors, offering a reliable approach in improving both flutter and VIV for bridges.

1. Introduction

1.1. Issues for bridge aerodynamic stability and necessity of mitigation measures

As science and technology advance, the improvement of bridge construction methods and usage of lightweight new materials allow bridges to be built with over long spans. Currently, the main span of the suspension bridge is reaching 2000 m, such as the Akashi Kaikyo Bridge (main span 1991 m), Japan, and Xihoumen Suspension Bridge (main span 1650 m), China. The main span of the cable-stayed bridge has exceeded 1000 m, such as the Russky Island Bridge (main span 1104 m), Russia, and Su Tong Yangtze River Bridge (main span 1088 m), China. With the increase of the bridge main span, the bridge structure becomes more flexible and has lower inherent damping, which makes the bridge more sensitive to wind loads and results in bridge flutter and vortex induced vibration (VIV) issues. Passive aerodynamic measure is the most effective and commonly used mitigation measure when a long-span bridge has a problem in fulfilling aerodynamic performance requirements (Wardlaw, 1972; Larsen, 1993; Larsen et al., 2000; Weight, 2009; Vaz et al., 2016; Yang et al., 2018). Systematical research on the vibration suppression effect and mechanism of different passive aerodynamic measures has been carried out (Jones et al., 1995; Saito and Sakata, 1999; Bruno and Mancini, 2002; Miyata, 2003; Wu and Kareem, 2012; Yang et al., 2015). To name a few, Sakai et al. (1993) compared different aerodynamic measures such as side plates, edge plates, flaps, fairing, baffle plates, and gratings, for possibly improving the bridge VIV; Irwin (2008) reviewed the vertical baffle plates installed under the deck to suppress the VIV of the Second Severn Bridge; Yang and Ge (2009) comparatively investigated the flutter control effects of aerodynamic measures, such as central stabilizers, central-slotting, adding fairings, adjusting the position of inspection rail, and horizontal stabilizers; and Fujino and Siringoringo (2013) reviewed the effective mitigation measures that have been used to avoid aerodynamic instability in the

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long-span bridges built all over the world. Yang et al. (2015) investigated the flutter stability of five typical box girder cross sections with various center slots, differing in the width, based on experimental and analytical analysis, and revealed the aerodynamic stabilization mechanism of the center slots by employing the particle image velocimetry (PIV) and proper orthogonal decomposition (POD) analysis.

1.2. Merits and limitations of the upward central stabilizer measure

Among various passive aerodynamic measures, central stabilizer has been favorably and successfully applied in many bridges for improving their flutter performance due to its specific merits (Miyata and Yamaguchi, 1993; Ge et al., 2009; Tang et al., 2018; Zhu et al., 2019; Chen et al., 2006). For example, Chen and Kareem (2006) explained the working mechanism of the central stabilizer in improving the flutter stability by investigating a closed box section and an open deck π-shaped section; and Ge and Xiang (2008) compared the flutter stability of two schemes, a narrowly slotted box girder with horizontal and central stabilizers and a widely slotted box girder, showing the superiority of using central stabilizers. Evidence shows that upward and downward central stabilizers with proper size can improve the flutter stability at positive and negative wind attack angles, respectively. The combined use of the upward and downward central stabilizers is significantly better than using each one alone regarding the vibration suppression effect. With the same central stabilizer, the upward one, i.e., placing on the deck, has a better effect on increasing the critical flutter wind velocity than that for the downward one, i.e., placing underneath the deck (Li et al., 2018).

Although the mitigation measure of the upward central stabilizers results in obvious beneficial effects on improving the flutter performance of bridge, in some cases, it is not suitable to be settled on the deck. For example, for a narrow bridge, setting the upward central stabilizers will hinder the vehicle on the bridge deck from turning around or switching the lane. When traffic jams or accidents occur, the setting of the upward central stabilizer is not conducive to accident rescue or ease of traffic pressure. Moreover, narrow bridges often encounter flutter instability problems. When other mitigation measures, such as the fairing, edge plate, and flaps cannot adequately improve the flutter stability of the bridge, it is necessary to find an appropriate alternative to the upward central stabilizer.

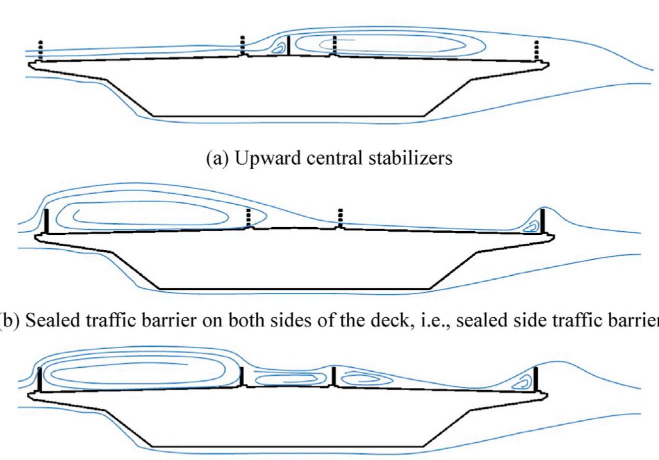
In addition, the mitigation measure of the upward central stabilizers could play negative roles in the attempts to control all the wind-induced effects of bridges, as sometimes it may cause the VIV of the bridge (Zhou et al., 2018). When the bridge designs encounter both flutter and VIV problems, adopting only one aerodynamic mitigation measure may have limited improvement in the aerodynamic stability of the bridge; in other words, a mitigation measure may be effective only for suppressing one kind of wind-induced vibrations. In this scenario, a combination of two or more aerodynamic measures to improve the aerodynamic stability of the bridge is thus highly recommended (Diana et al., 2013; Larsen and Larose, 2015; Yang et al., 2017). However, it is desirable to use only one aerodynamic measure to effectively improve all the aspects of the aerodynamic stability of the bridge, i.e., both the flutter and VIV of most bridges. Therefore, it is essential to find an alternative mitigation measure such that it can play the role, similar to the upward central stabilizers, in improving the flutter stability of the bridge without incurring the VIV of the bridge.

1.3. Proposal of an alternative measure to the upward central stabilizer

To propose a desirable alternative to the upward central stabilizer measure, two guidelines should be followed, using as few aerodynamic measures as possible and at the same time, resulting in no significant additional structure costs or design. According to previous studies on the wind-induced vibration, the overall cross-sectional shape of the bridge deck can be a crucial factor for affecting the wind-induced vibration of the bridge (Larsen, 1998, 2008; Matsumoto et al., 1999; Sato et al., 2000; Fujino and Yoshida, 2002; Larsen and Wall, 2012). The aerodynamic shape of the bridge can lead to major perturbations of the flow pattern, directly determining the positions of the separation point, reattachment, wake and the formation of the vortex. Railings, median dividers, border beams can modify the flow pattern and influence the aerodynamic stability of the bridge (Wang et al., 2009). Nagao et al. (1995, 1997) have observed that the shape and position of the traffic barriers have significant effects on VIV. It is found that the intensity of the exciting force and the shedding of vortices largely depend on whether the traffic barriers were directly hit by the shear layer or were immersed on the separation bubble. Kubo et al. (2001, 2002) also found the solid traffic barrier location and attack angle have significant impacts on the aerodynamic responses of the π section due to the interference effect of the separated flows. The contribution from the separation bubble generated on the upper side of the π section was dominant for the torsional motion.

Since the traffic barrier, sidewalk traffic barrier and upward central stabilizer are all located above the deck, when it is not suitable for setting the mitigation measure of the upward central stabilizer, sealing the traffic barrier or sidewalk traffic barrier would be a potential choice. These measures can change the aerodynamic shape of the bridge deck (Li et al., 2018). Fig. 1 shows the possible flow field around the box girder section with different mitigation measures. It can be expected that the flow field below the bridge deck of the sealed traffic barrier or sidewalk traffic barrier is basically the same as that with the upward central stabilizer. In Fig. 1(a), with the presence of the upward central stabilizer, large-scale vortices would occur on the downstream side of the bridge deck. With the sealed the traffic barrier on both sides of the deck, large-scale vortices would appear on the upstream side of the bridge deck, as shown in Fig. 1(b). In Fig. 1(c), with the sealed side and central traffic barriers, large-scale vortices would appear on both the upstream and downstream sides of the bridge deck, and the vortices on the upstream side would be larger than that on the downstream side. Note that large-scale vortices dominate the wind-induced vibration of bridge. Because sealed bridge deck traffic barrier or sidewalk traffic barrier may produce large-scale vortices on the deck surface, they may offer a vibration suppression effect on the bridge flutter, similar to that of the upward central stabilizer.

Note that sealing the traffic barrier or sidewalk traffic barrier on the bridge deck may cause VIV of the bridge. To avoid this, one effective method is to change the aerodynamic shape and prevent periodic vortex shedding on the surface of the bridge. Fig. 2 shows the sketched 3D flow field associated with the sealed traffic barrier or partially sealed traffic barrier on the deck. It can be expected from Fig. 2(c) that for the sealed traffic barrier, the flow pattern of the incoming flow would be consistent



(a) Upward central stabilizers

(b) Sealed traffic barrier on both sides of the deck, i.e., sealed side traffic barrier

(c) Sealed side and central traffic barriers

Fig. 1. Sketched flow field around the box girder section with different mitigation measures.

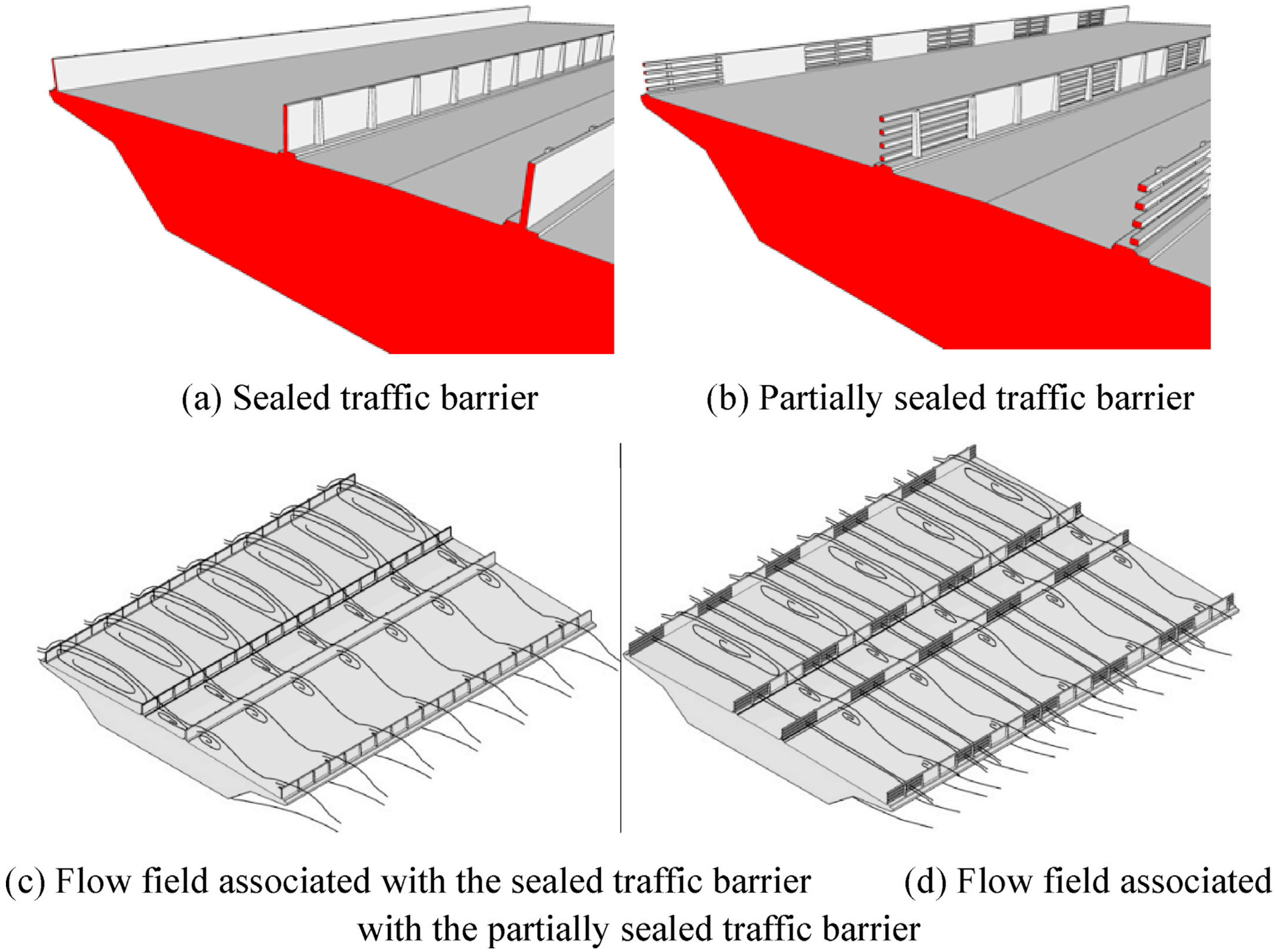


Fig. 2. Sketched flow field associated with the sealed or partially sealed traffic barriers on the deck.

along the length of the bridge. When periodic vortex shedding exists on the bridge surface, the energy of the aerodynamic force is relatively large, thus probably inducing the VIV. In Fig. 2(d), there should be a significant difference between the flow field passing through the sealed zone and that for the non-sealed zone. Therefore, the way of partially sealed traffic barrier or sidewalk traffic barrier would make the flow pattern of the incoming flow to change continuously along the length of the bridge. As a result, this could disturb the periodic vortex shedding on the bridge surface or weaken the aerodynamic force energy of the periodic vortex shedding, thereby suppressing the vortex-induced vibration.

1.4. Objective of the present study

As discussed above, the objective of the present study is to propose an alternative mitigation measure, i.e., sealed traffic barrier, to the upward central stabilizer and verifies its efficacy in improving the flutter and vortex induced vibration stability of bridges through wind tunnel tests. The working mechanism that the (partially) sealed traffic barrier or sidewalk traffic barrier interferes with the occurrence of the periodic vortex shedding on the bridge surface to suppress the VIV will be studied. Three distinctive real-world bridge projects, the Hongkong-Zhuhai-Macao Bridge, Liujiaxia Bridge, and Chongqing Hechuan Bridge, will be involved in the procedure.

The remainder of the study is organized as follows:Firstly, the engineering background for the three typical bridge projects is presented. Then, the aerodynamic performance of the Hongkong-Zhuhai-Macao Bridge, Liujiaxia Bridge, and Chongqing Hechuan Bridge is sequentially studied through the wind tunnel test, where the bridge VIV and/or flutter is suppressed by the partially sealed side traffic barrier or sidewalk traffic barrier. As follows, the effect of the sealed form for the traffic barrier on the bridge VIV and flutter is studied. Concluding remarks are given finally. It is hoped that this proposed measure can be taken as an alternative aerodynamic measure for bridge engineering.

2. Engineering background

In this study, three typical real-world bridge projects, the Hongkong-Zhuhai-Macao Bridge, Liujiaxia Bridge, and Chongqing Hechuan Bridge, conducted by the leading authors in the wind tunnel tests, are selected to demonstrate the efficacy of the proposed sealed barrier measure on suppressing the wind-induced vibration of the bridge. The reasons behind this selection are that: (1) Their main beam types are representative and commonly used in long-span bridges, where the Hongkong-Zhuhai-Macao Bridge adopts a bluff body box girder, Liujiaxia Bridge uses a steel truss girder, and Chongqing Hechuan Bridge employs a streamlined slotted box girder; (2) The original designs for all the three bridges have problems associated with aerodynamic instability, where the VIV was observed for the Hongkong-Zhuhai-Macao Bridge, the flutter was observed for the Liujiaxia Bridge, and both the VIV and flutter were observed for the Chongqing Hechuan Bridge. These vibrations are also the main forms of the wind-induced vibration for bridges; and (3) It is highly challenging to effectively improve the aerodynamic instability performance for these bridges by considering common aerodynamic measures. Therefore, the variants of the sealed traffic barrier or sidewalk traffic barrier measure are proposed and conducted, hoping to solve the aerodynamic instability issues for these bridges. The introduction of these three bridges is given as follows.

2.1. Hongkong-Zhuhai-Macao bridge

The Hongkong-Zhuhai-Macao Jianghai Zhidachuan Waterway Bridge, shorted as the Hongkong-Zhuhai-Macao Bridge here, is a cable stayed bridge with three towers and its span layout is 110 + 129+258 + 258+129 + 110 m , with a total length of 994 m. The bridge adopts three steel towers with heights of 109.8 m and 108.5 m for the central and side towers, respectively. Its stiffening girder is a bluff body steel box girder. The elevation view and box girder section of the Hongkong-Zhuhai-Macao Bridge are shown in Figs. 3 and 4, respectively. The bridge site is featured with a complicated, volatile, frequent disastrous weather, especially the adverse windy conditions induced by the tropical cyclones (typhoons). Besides, under the influence of the tropical climate, the near ground surface wind, formed by the unique complex topography of the land, will also have an important impact on the wind resistance design of the Hong Kong-Zhuhai-Macao Bridge.

2.2. Liujiaxia Bridge

The Liujiaxia Bridge is a suspension bridge spanning the branch of the Liujiaxia Reservoir and its span layout is 20 + 536+ 20 m, as shown in Fig. 5. Its stiffening girder is a steel truss girder with two-way lanes, 11.0 m for the total width, as shown in Fig. 6. In the completion state of the bridge, the sag-span ratio is about 1:11, the center-to-center distance of the main cable is 15.6 m. The pylon is designed as Islamic architectural style concrete-filled steel tube tower with the column diameter of 2.8 m. Because of the narrow bridge deck, its aerodynamic stability needs to be assessed and improved.

2.3. Chongqing Hechuan Bridge

The Chongqing Hechuan Bridge is a ground-anchored suspension bridge with the main span of 400 m, as shown in Fig. 7. The bridge adopts a single main cable and mountain-shaped pylons. The slotted box girder consists of two separable steel box structures with variable intervals, connected by cross beams, along the longitudinal direction. The width of one single steel box is 8.5 m, and the interval gap between the steel boxes varies from 2 m at the middle span to 16 m at the pylon. The plan view of the girder presents an X-shape. Fig. 8 shows the elevation and plan views and box girder section for this bridge.

3. Improving the VIV performance for Hongkong-Zhuhai-Macao bridge

3.1. Experimental set-up for Hongkong-Zhuhai-Macao bridge

For the Hongkong-Zhuhai-Macao Bridge, its flutter performance can meet the requirements of the wind-resistant specification for highway bridges of China (Ministry of Communications of the People's Republic of China, 2004) through the wind tunnel test. Therefore, investigating its VIV performance will be the focus here. For this purpose, two section models with scales 1:50 and 1:20 are sequentially tested in the wind tunnel, as shown in Fig. 9, where the larger one is specifically used to verify the VIV performance and vibration suppression effect of the proposed measure. Table 1 lists the parameters for the section models with scales 1:50 and 1:20 and Table 2 lists the test conditions in the wind tunnel.

Experiments were conducted in a closed circuit wind tunnel with the working section size of 2.5 m in height, 3 m in width, and 15 m in length. The wind speed can be adjusted continuously from 2 m/s to 53 m/s in the test section, with the turbulence intensity lower than 0.3% in smooth flow. The section model was supported by eight springs, allowing heaving and torsional vibrations. The model consists of steel and aluminum skeleton, lined with balsa wood. Two plates were fixed at both ends of the model to eliminate the boundary effects of the supporting system.

3.2. VIV performance of section model for Hongkong-Zhuhai-Macao bridge

Fig. 10 shows the vertical and torsional VIV responses of the section model with scale 1:50 for the Hongkong-Zhuhai-Macao Bridge at wind attack angles -5°, -3°, 0°, +3°and +5°. Note that the maximum values of the responses and the wind speed have been conformed to the prototype values. The vertical VIVs were observed at wind attack angle +3°and +5°. The torsional VIVs were observed at wind attack angles -5°,-3°and 0°. The maximum vertical VIV amplitude is 0.384 m, which occurs at the wind attack angle +5°with the wind speed 26.0 m/s. The length of the vertical VIV lock-in district at wind attack angle +5° is relatively large, from 20.5 m/s to 31.5 m/s. The vertical VIVs were also observed at wind attack angles +3° and 0°, but the maximum amplitude and the length of the lock-in district were much smaller than that at the wind attack angle +5°. The torsional VIVs were observed at the negative wind attack angle. The maximum values of the torsional VIV at wind attack angles -5°and -3° are 0.248°and 0.284°, respectively, and the corresponding wind speeds are 23.5 m/s, within the torsional VIV lock-in district from 23 m/s to 24.5 m/s. The torsional VIV also observed at wind attack angle 0° in this lock-in district, but the maximum amplitude is smaller, which is 0.096°, than that at attack angles -5°and -3°.

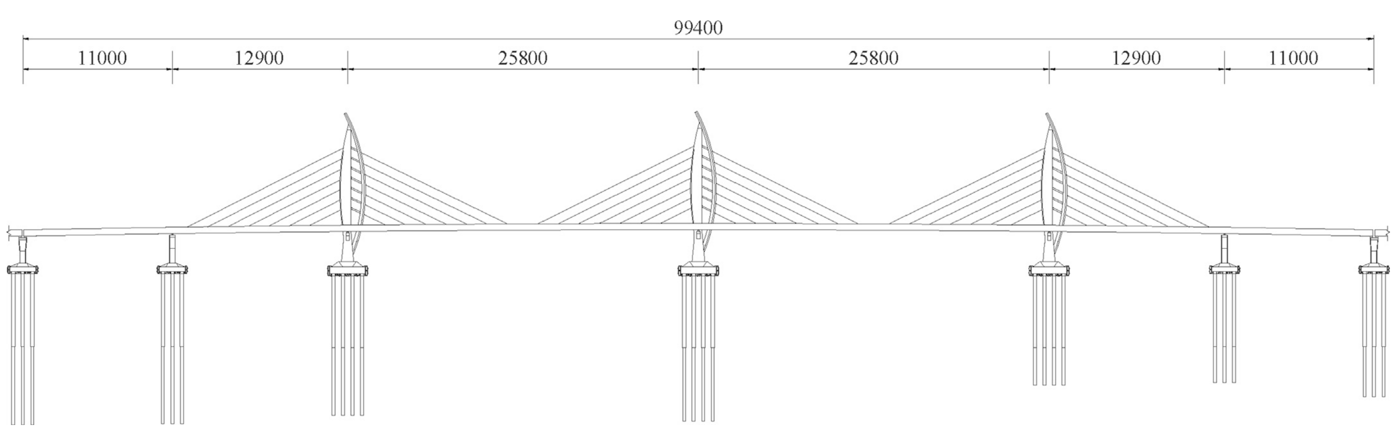


Fig. 3. Elevation view of the Hongkong-Zhuhai-Macao Bridge (Unit: cm).

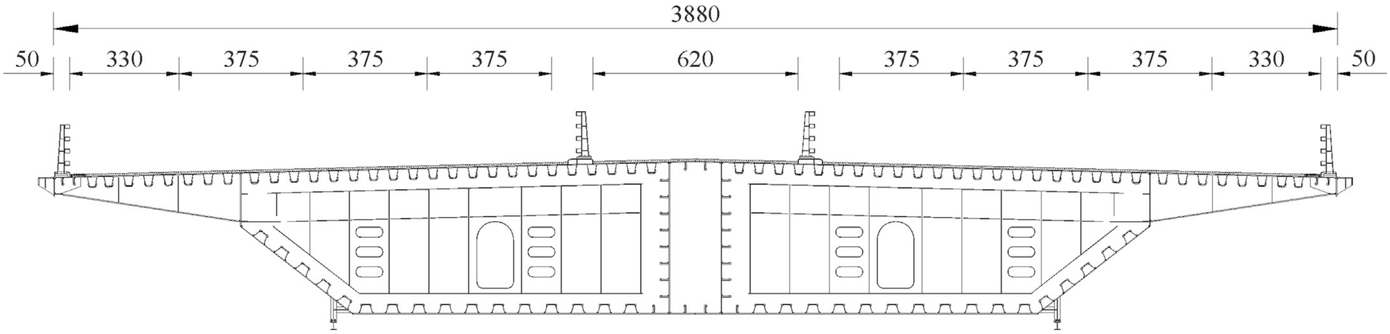


Fig. 4. Box girder section of the Hongkong-Zhuhai-Macao Bridge (Unit: cm).

Fig. 5. Rendered view of the Liujiaxia Bridge.

Based on the wind-resistant design specification for highway bridges of China (Ministry of Communications of the People's Republic of China, 2004), the amplitude limitations for vertical and torsional VIVs ($[h_\alpha]$and $[\theta_\alpha]$) are correspondingly defined as follows:

Fig. 7. Rendered view of the Chongqing Hechuan Bridge.

$$[h_\alpha] = 0.04/f_h = 0.119 \quad m \tag{1}$$

$$[\theta_\alpha] = 4.56/(Bf_t) = 0.115^\circ \tag{2}$$

where f_h, f_t and B are the vertical frequency, torsional frequency, and width of the section model with scale 1:50 for the Hongkong-Zhuhai-Macao Bridge (see Table .1). Compared with the VIV amplitude limitations in the Chinese specification, the maximum amplitudes of the

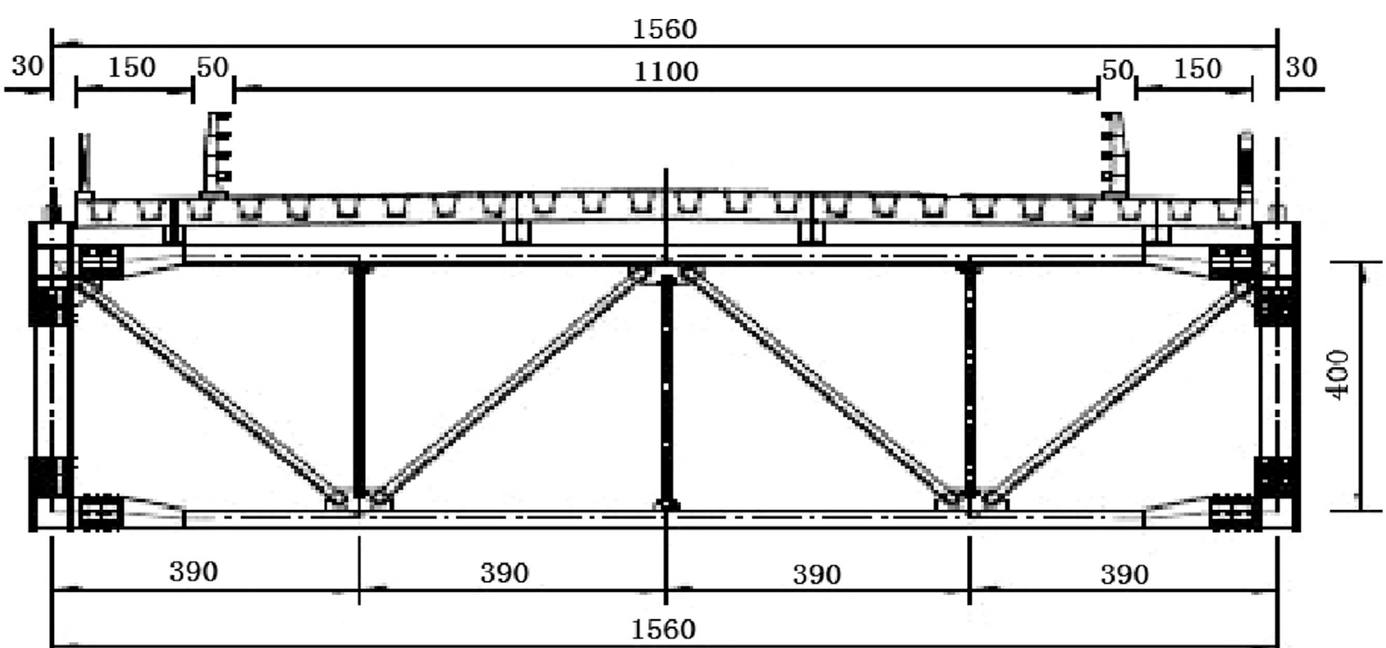
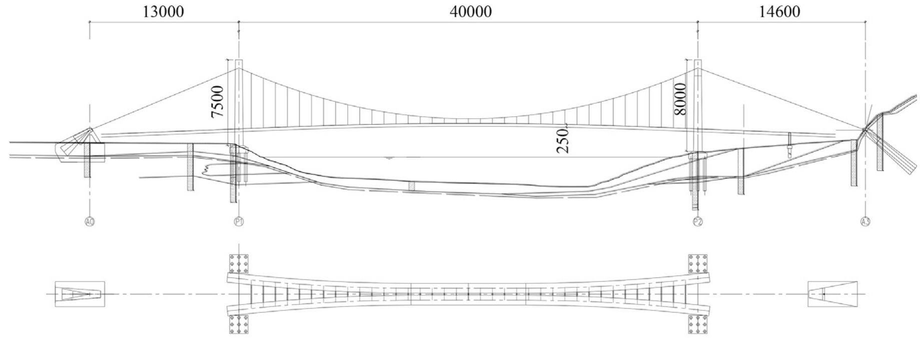
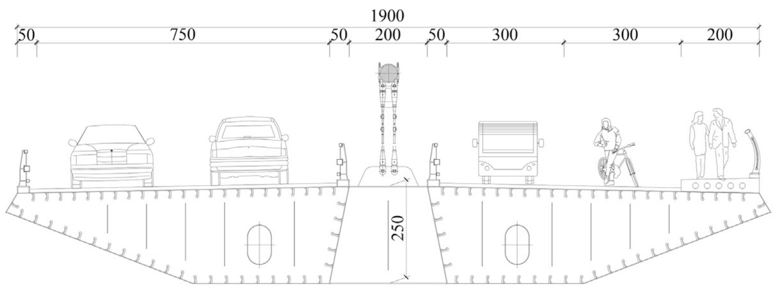


Fig. 6. Steel truss stiffening girder of the Liujiaxia Bridge (unit: cm).



(a) Elevation and plan (Unit: cm)



(b) Box girder section (Unit: cm)

Fig. 8. Schematic description of the Chongqing Hechuan Bridge.

(a) 1:50 scale section model (b) 1:20 large-scale section model

Fig. 9. Section models for the Hongkong-Zhuhai-Macao Bridge in the wind tunnel.

vertical VIV at wind attack angle $+5^{\circ}$ and the torsional VIVs at wind attack angles $-5^{\circ}$and -3° cannot satisfy the design requirements. Therefore, effective aerodynamic measures should be used to improve the VIV performance of the Hongkong-Zhuhai-Macao Bridge.

3.3. Sealed traffic barrier mitigation measure for Hongkong-Zhuhai-Macao bridge

To suppress the VIVs for the Hongkong-Zhuhai-Macao Bridge, a series of vibration suppression measures have been investigated, including changing the shape of the fairing and inspection vehicle rail, and setting horizontal guide plate and suppressing plates. However, the vibration suppression effects for the above measures were not good. Therefore, the sealed traffic barrier was configured on both sides of the bridge deck model, aiming to improve the aerodynamic stability. Here, two cases were considered, one-seal-three-empty barrier (Measure 1) and one-seal-two-empty barrier (Measure 2), with ventilation rates being 75% and 67%, respectively, as shown in Fig. 11.

The vertical and torsional responses of the 1:50 section model for the Hongkong-Zhuhai-Macao Bridge with considering Measures 1 and 2 at wind attack angles -5°, -3°, 0°, $+3^{\circ}$and $+5^{\circ}$ are shown in Figs. 12 and 13, respectively. It can be seen that the VIV can be effectively suppressed by

Table 1
Parameters for section models of the Hongkong-Zhuhai-Macao Bridge in the VIV test.

Properties	Parameter	Unit	Prototype	Section model (Scale ratio)	
				1:50	1:20
Width	B	M	38.8	0.776	1.94
Mass per unit length m	M	kg/m	35000	14.00	87.50
Mass moment of inertia per unit length	J_m	kg.m^2/m	3630000	0.58	22.69
Vertical frequency	f_b	Hz	0.307	6.75	3.37
Torsional frequency	f_t	Hz	0.631	20.43	10.21
Vertical damping ratio	ξ_b	%	0.5	0.25	0.36
Torsional damping ratio	ξ_t	%	0.5	0.27	0.33

Table 2
Test conditions for section models of the Hongkong-Zhuhai-Macao Bridge.

Section model (Scale ratio)	Wind attack angle (°)	Wind speed ratio	Flow field
1:50	−5, −3, 0, +3, +5	1:2.5	Uniform oncoming flow
1:20	−5, −3, 0, +3, +5	1:2.0	Uniform oncoming flow

sealing the traffic barrier with regular intervals on both sides of the deck along the bridge. The large amplitude value for the vertical VIV of the original section model within the wind speed lock-in district from 20.5 m/s to 31.5 m/s at wind attack angle +5° was successfully suppressed by Measures 1 and 2. In addition, both the amplitude value and the length of the wind speed lock-in district were substantially reduced. The small amplitude value for the VIV at wind attack angles 0° and +3° were completely eliminated by the action of Measures 1 and 2. However, the vibration suppression effect of sealing the traffic barrier with regular intervals on both sides of the bridge deck on the torsional VIV was not as obvious as that on the vertical VIV. Specifically, torsional VIVs, exceeding the limitations defined in the specification, were still observed for the segment model with Measure 1 at wind attack angles -5° and -3°. The VIV lock-in district did not change much, but the maximum amplitude of the torsional VIV was significantly reduced compared to that of the original section.

For Measure 2, desirable vibration suppression effects have achieved, where the maximum amplitude of the torsional VIV was largely reduced, thereby meeting the requirements by the specification. It can be seen that the sealed traffic barrier with proper vertical intervals can effectively suppress the VIVs of the bridge, and the vibration suppression effect on the vertical VIVs are obvious. The comparisons between the results associated with Measures 1 and 2 show that the sealed traffic barrier with an appropriate smaller ventilation rate can result in better suppression

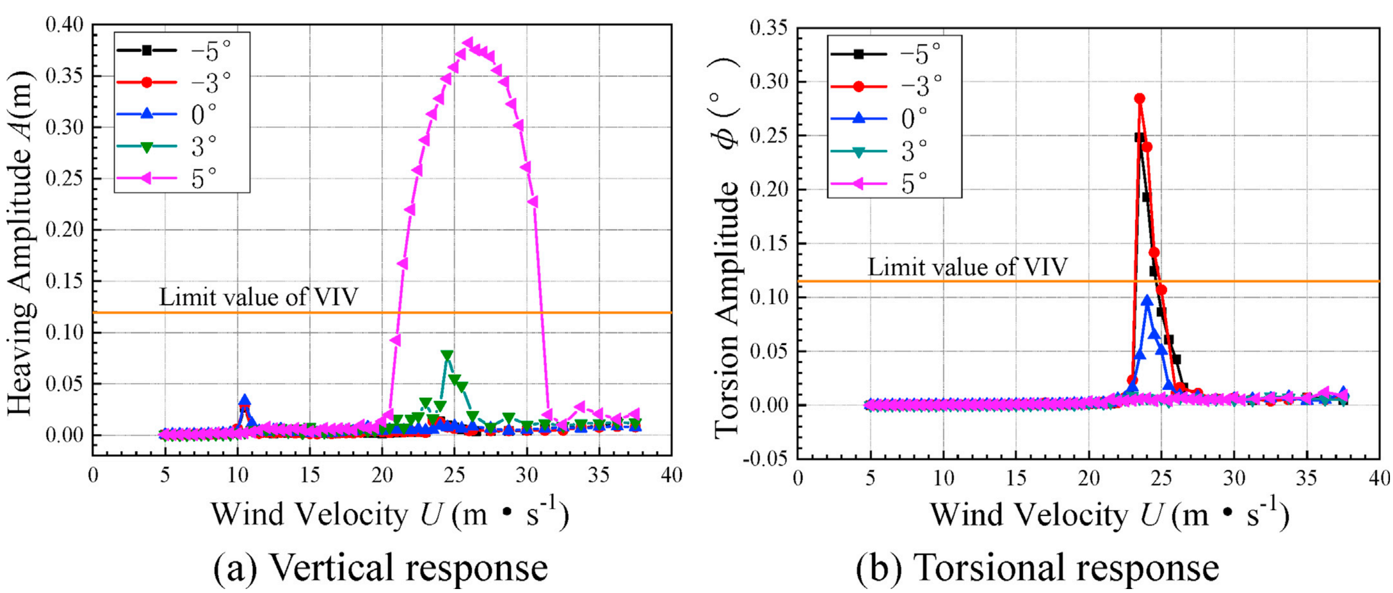


Fig. 10. Vertical and torsional VIV responses of the section model with scale 1:50 for the Hongkong-Zhuhai-Macao Bridge.

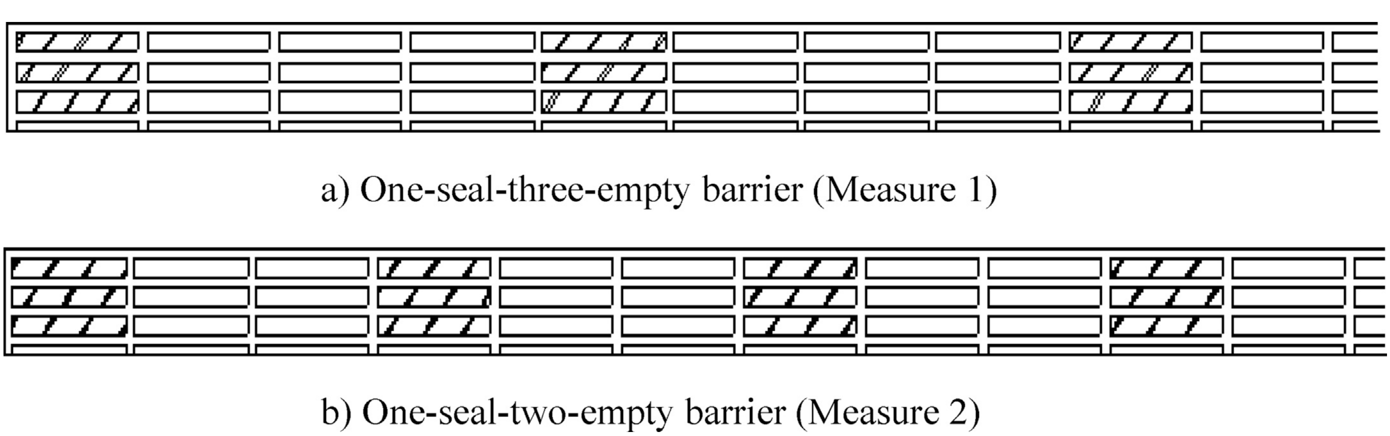
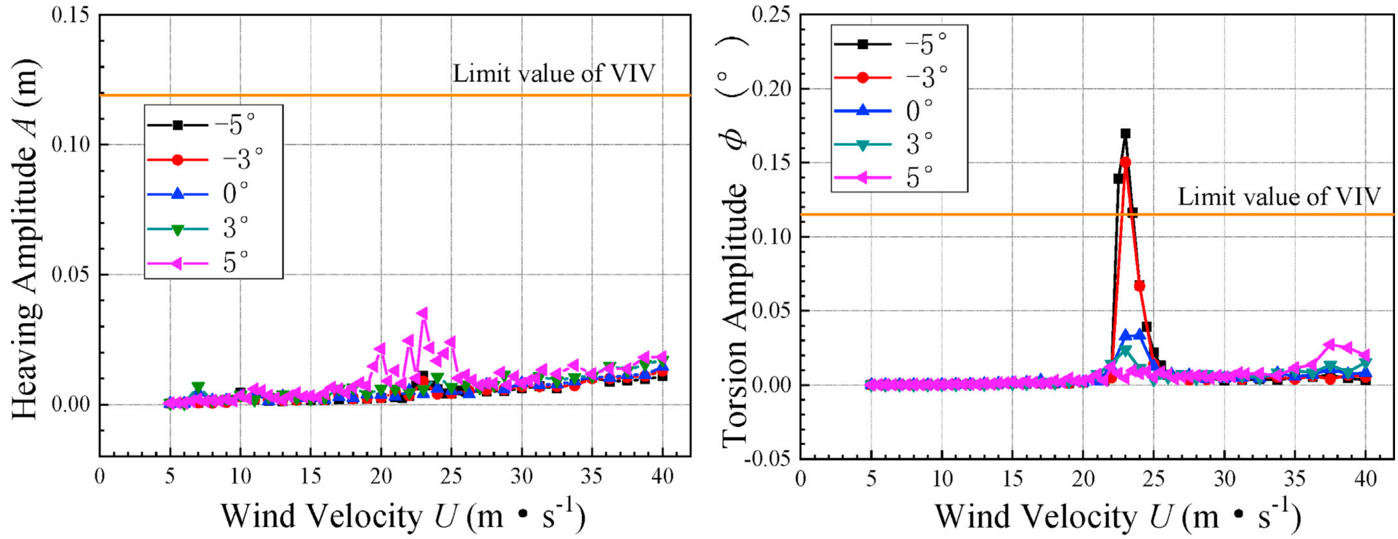
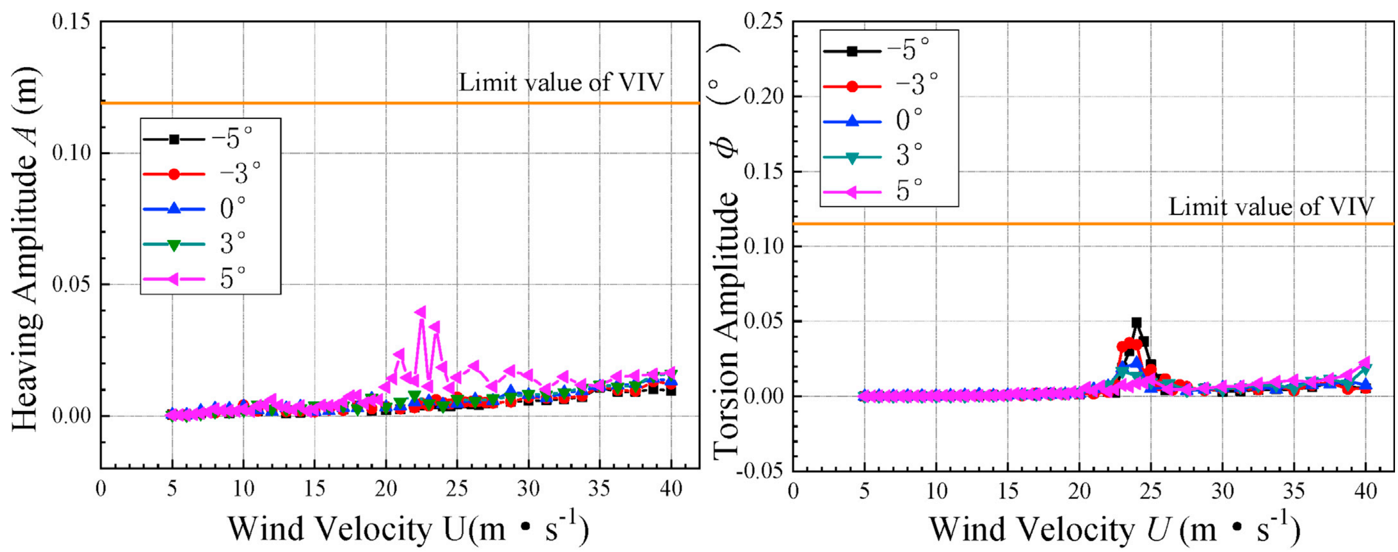


Fig. 11. Suppression measures for the Hongkong-Zhuhai-Macao bridge.



(a) Vertical response (b) Torsional response

Fig. 12. Vertical and torsional VIV responses for Measure 1.



(a) Vertical response (b) Torsional response

Fig. 13. Vertical and torsional VIV responses for Measure 2.

effects for the VIV under considered wind attack angles, while one measure with a larger ventilation rate may not perform favorably well. As such, it is expected that once the interval distance of the sealed traffic barrier is appropriate, the effectively vibration suppression effect on the torsional VIV can also be achieved.

3.4. Verification by using section model with scale 1:20

The large-scale section model test verifies the efficacy of the sealed traffic barrier on suppressing the VIVs observed in conventional section model vibration test of the Hongkong-Zhuhai-Macao Bridge. The vibration tests for the 1:50 section model with considering three aerodynamic measures (changing the shape of the fairing and inspection vehicle rail, setting horizontal guide plate and suppressing plates, and sealing the traffic barrier) show that the one-seal two-empty barrier (Measure 2) can effectively control the VIV amplitude within the specified limit.

The vertical and torsional responses of the 1:20 large-scale section model for the Hongkong-Zhuhai-Macao Bridge with considering the aerodynamic Measure 2 at wind attack angles -5°, -3°, 0°, +3°and +5° are shown in Fig. 14. It can be seen that the use of Measure 2, i.e., sealing the traffic barrier with appropriate vertical intervals, can successfully suppress the VIV of the original section. The geometric format of Measure 2, one-seal-two-empty barrier, allows it to be simply set up, without significant additional structure costs.

To consider the Reynolds number effects due to the scale of the section model, Fig. 15 specifically shows the vertical and torsional responses of both section models with Measure 2 configured for the Hongkong-Zhuhai-Macao Bridge at wind attack angles -3°, 0°, and +3°. It is noteworthy that: (1) for the vertical responses, these two section models follow the same trend in amplitude when the wind speed is below 25 m/

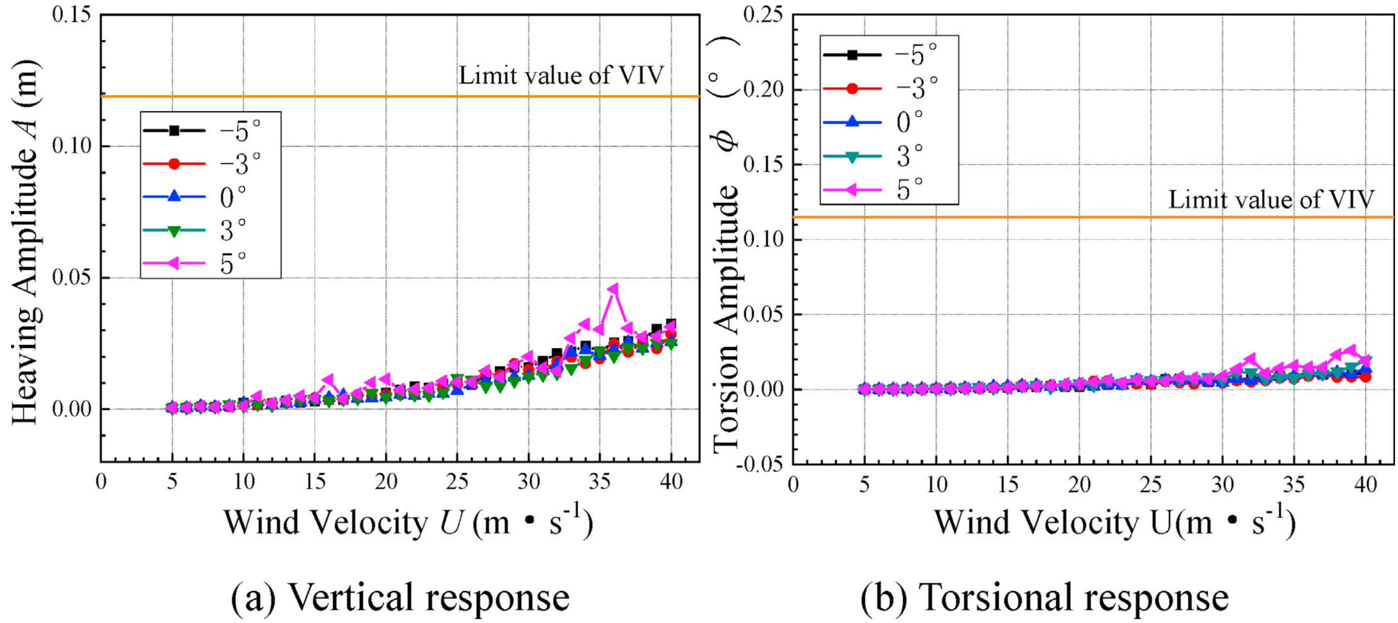


Fig. 14. Vertical and torsional VIV responses associated with Measure 2 for the large-scale section model wind tunnel test.

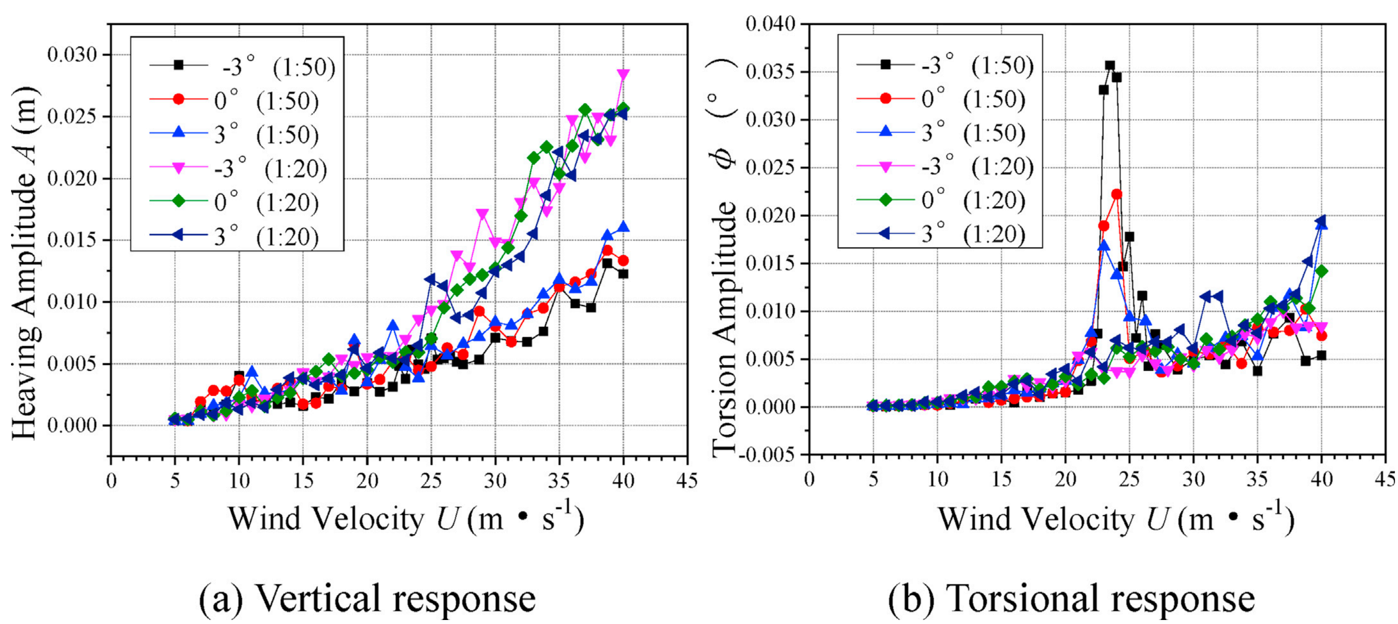


Fig. 15. Reynolds number effects for the section model with Measure 2.

s, whereas when the wind speed is above 25 m/s, the heaving amplitude for the 1:20 large-scale section model is relatively larger than that for the 1:50 scale model. However, the vertical VIV is not observed on both section models; (2) for the torsional responses, the section models follow similar trend on the torsion amplitude. However, a small VIV is observed within the wind speed range from 20.5 m/s to 31.5 m/s for the 1:50 scale model, but its amplitude is much smaller than the threshold in the code; and (3) section models with conventional scales are usually adopted to evaluate the vibration performance for bridges in wind tunnel tests. For long-span bridges, a large-scale section model can be used for verification purpose with permitted economic and time conditions.

3.5. Discussion on mitigation mechanism of proposed measure

Computational Fluid Dynamic (CFD) technique is used here to illustrate the suppression mechanism of the sealed traffic barrier on the Hongkong-Zhuhai-Macao Bridge. The computational domain and boundary conditions are given in Fig. 16, where the section model is placed in the specified position with wall boundaries set along its surface. Unstructured mesh is considered in the whole computational domain, similar to the mesh shown in Fig. 18 later, and the y + value for the bridge deck ranges from 1.0 to 10.0. The shear stress transport (SST) *k-ω* model is adopted as the turbulence closure for the RANS (Reynolds-averaged Navier-Stokes) equations, the SIMPLEC (Semi-Implicit Method for Pressure Linked Equations-Consistent) scheme is used as the solver for the pressure-velocity coupling method, and second-order upwind is used for momentum advection terms. The wind profile considered at the inlet boundary is a uniform flow with speed 10 m/s.

To verify the accuracy of the numerical simulations, the 1:50 scale section model without any mitigation measure is placed in the

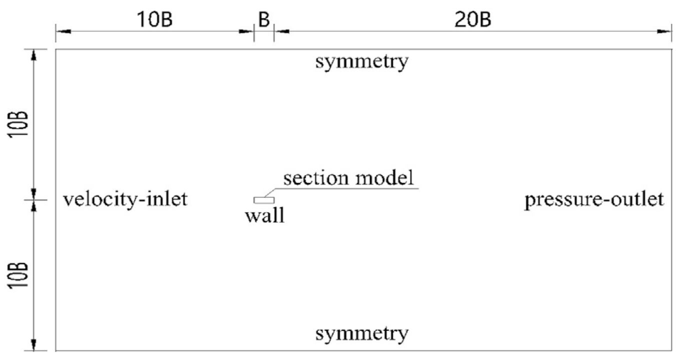


Fig. 16. Computational domain and boundary conditions.

computational domain and the comparison between the aerodynamic coefficients from the wind tunnel tests and those via numerical simulations is shown in Fig. 17. It is noted that the drag and moment coefficients between these two approaches agree with each other favorably well. However, for the lift coefficients, the results by the numerical simulations is relatively smaller than that from the wind tunnel tests under wind attack angles from −3° to 10°, whereas the results by these two approaches follow the general development trend. Therefore, numerical simulations are further carried out to reasonably explain, though not with high fidelity, the mitigation mechanism of the proposed measure.

As follows, bridge section models with three distinctive mitigation configurations, the upward central stabilizer, sealed (side) traffic barrier, and partially sealed (side and central) traffic barrier, are placed in the computational domain and the corresponding mesh grids are demonstrated in Fig. 18. Again, unstructured mesh is adopted in the computational domain. Note that since these simulations are two dimensional (2D), the ventilation rate in the case with the partially sealed traffic barrier is equivalently treated following the “two-seal-two-empty” scheme, with its value being 50%, which is different from Measures 1 and 2 as used in the wind tunnel tests.

Fig. 19 shows the flow field around the box girder section models with three distinctive mitigation configurations for the Hongkong-Zhuhai-Macao Bridge by using numerical simulations. Apparently, these measures can modify the aerodynamic shape of the bridge deck. It is worth noting in this figure that: (1) The flow field below the bridge deck in the cases with sealed or partially sealed traffic barrier is basically

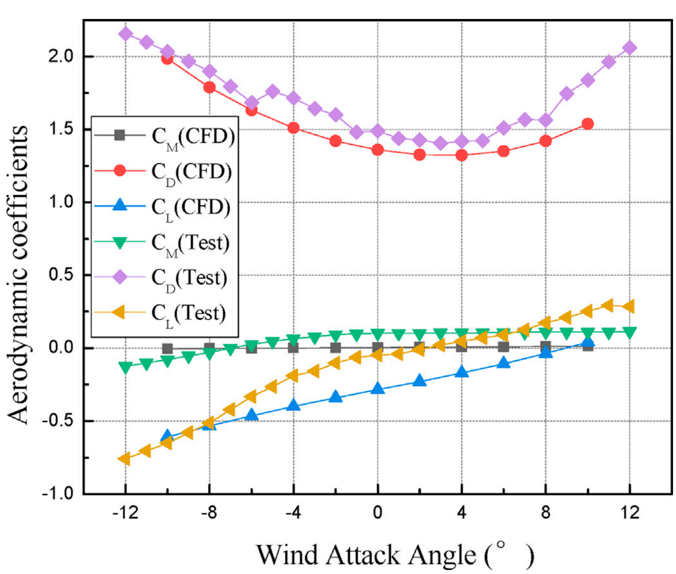


Fig. 17. Comparison of the aerodynamic coefficients for the section model of the Hongkong-Zhuhai-Macao Bridge between wind tunnel tests and numerical simulations.

the same as that associated with the upward central stabilizer, whereas a larger-scale vortex is well formed at the rear part of the bridge in the case with the sealed barrier and the vortex in the case with the partially sealed barrier is in the formation process. Although these mitigation measures considered are all located above the deck, posing less effect on the flow field below the bridge deck, they would introduce slightly different aerodynamic effects on the formation of vortices at the rear part of the bridge; (2) In the case with the upward central stabilizer, large-scale vortices occur at the downstream side (since the position of the stabilizer) above the bridge deck. Similar phenomenon is noted in the case with partially sealed traffic barrier, but with slight difference for the location of the vortices. This similarity explains that partially sealed traffic barrier can produce notable vibration suppression effect as that of the upward central stabilizer; and (3) In the case with sealed traffic barrier, a notable large-scale vortex appears at the upstream side of the deck. As large-scale vortices dominate the wind-induced vibration for the bridge, the modification of the field flow by the proposed sealed traffic barrier measure can successfully suppress the VIV, similar to the upward central stabilizer. Thus, sealing the traffic barrier appropriately can serve as a competitive alternative to the upward central stabilizer.

Three dimensional (3D) section models are constructed to purposely demonstrate, though not with high fidelity, the features of the flow field around the bridge deck and the results for cases with sealed and partially sealed traffic barriers are shown in Fig. 20. It is observed in Fig. 20(a) that the flow pattern of the flow field is consistent along the length of the bridge model and periodic vortex shedding is produced at the bridge deck surface, largely resulting in the VIV phenomenon. However, in the case with partially sealed traffic barrier, there is a significant difference between the air flow passing through the sealed zone and that passing through the non-sealed zone. Therefore, the partially sealed traffic barrier alters the flow pattern along the length of the bridge and disturbs the periodic vortex shedding on the bridge surface, thus weakening the aerodynamic force energy of the periodic vortex shedding and suppressing the vortex-induced vibration.

Fig. 21 shows the aerodynamic coefficients for the bridge model with considering the aforementioned three distinctive mitigation measures based on 2D numerical simulations. As noted in this figure, sealing the side traffic barriers on the deck results in large wind resistance and therefore, the section model with the sealed traffic barrier bears notable larger drag, lift, and moment coefficients under most wind attack angles than the other two cases. In contrast, the differences between the aerodynamic coefficients, especially C_M, associated with the upward central stabilizer and partially sealed traffic barrier are relatively small; the reason lies in their similar aerodynamic flow field around the bridge deck.

4. Improving the flutter performance for Liujiaxia Bridge

4.1. Experimental set-up for Liujiaxia Bridge

The VIV stability of the Liujiaxia Bridge model can meet the requirements of the Chinese wind-resistant design specification through the wind tunnel test, whereas the flutter stability is an issue that needs to be addressed. To this end, a section model with scale 1:40 was tested in the wind tunnel. Fig. 22 shows the test setup for the Liujiaxia Bridge model. Table 4 lists the parameters for the flutter test of the Liujiaxia Bridge and Table 5 lists the test conditions.

4.2. Flutter performance of Liujiaxia Bridge

Table 6 shows the flutter critical wind speed for the Liujiaxia Bridge at different wind attack angles in the uniform flow field. After conforming the test wind speed to the prototype values, the flutter critical wind speeds for the Liujiaxia Bridge at wind attack angles -3°, 0°and +3° are 48.2 m/s,43.8 m/s and 41.6 m/s, respectively.

Based on the wind-resistant design specification for highway bridges

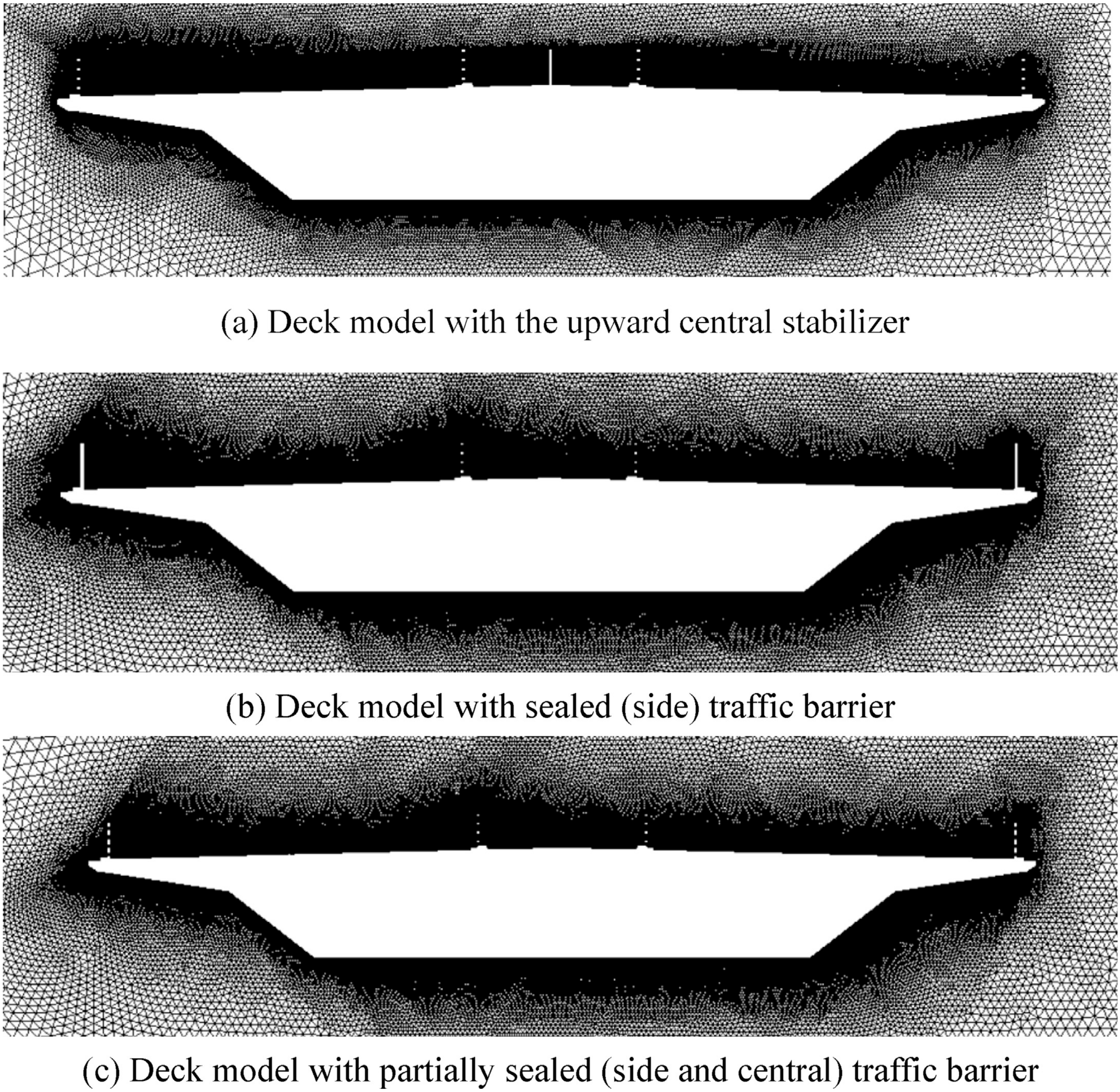


Fig. 18. Mesh grids for bridge section models with three distinctive mitigation measures.

(Ministry of Communications of the People's Republic of China, 2004), the flutter checking wind speed ($[V_{cr}]$) is defined as follows:

$$[V_{cr}] = 1.2\mu_f V_d = 51.1 \quad m/s \tag{3}$$

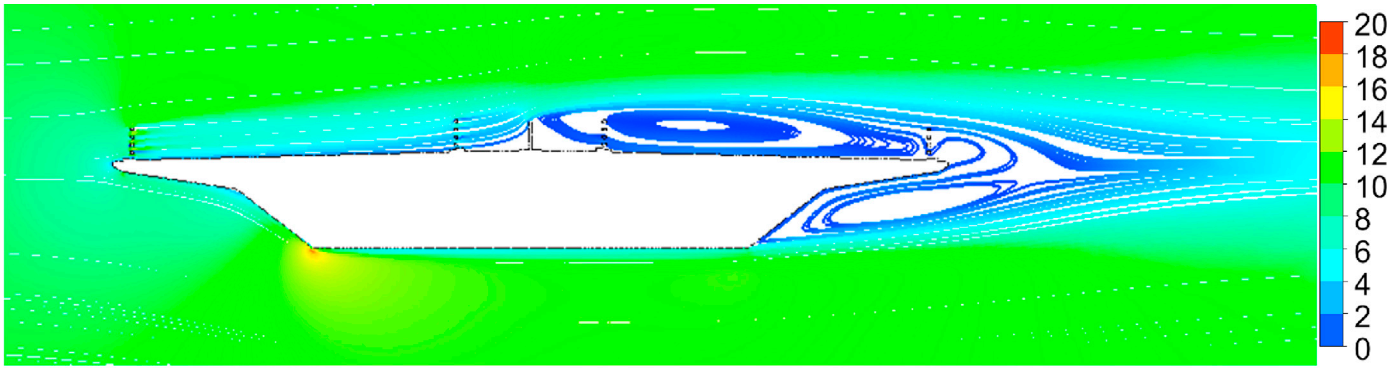
where $\mu_f = 1.23$ is the fluctuation wind speed correction factor and $V_d =$ 34.6 m/s is the designed standard wind speed for the Liujiaxia Bridge.

The results show that the flutter stability for the Liujiaxia Bridge does not meet the requirements of the Chinese wind-resistant design specification at wind attack angles -3°,0° and +3°. Therefore, aerodynamic measures need to be taken to improve its flutter stability. As follows, three aerodynamic measures, the central stabilizer, guide plate, and the sealed side traffic barrier, were tested.

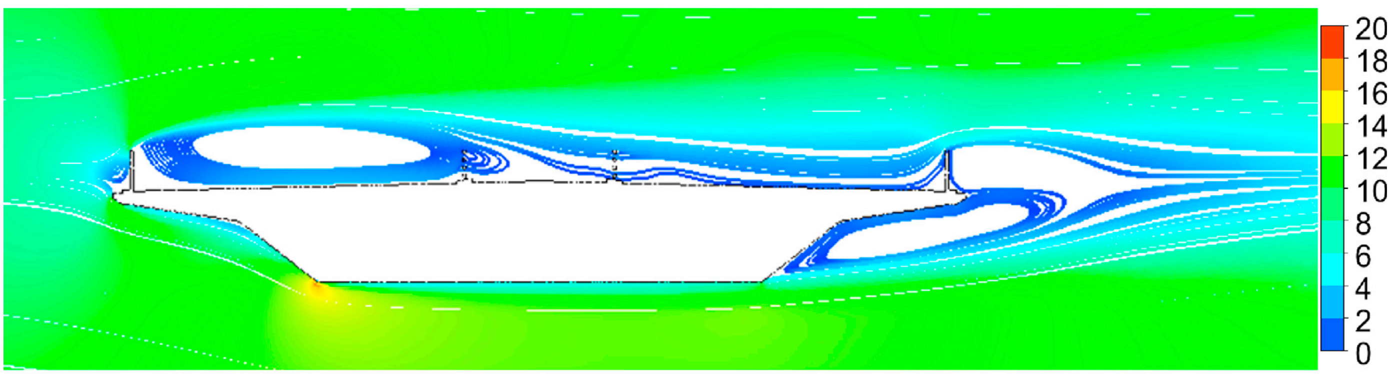
4.3. Central stabilizer measure for Liujiaxia Bridge

It is widely known that the central stabilizer is an effective measure to improve the flutter stability of the bridge. The installation of the upper and downward central stabilizers should take into account factors such as the traffic situation on the bridge deck, construction feasibility, and aesthetics. Here four variants of the central stabilizer combination measure, named Cases 1 to 4, are selected to investigate their influence on the flutter stability performance of the Liujiaxia Bridge, and the settings of these variants are given in Table 7. Table 8 shows the flutter critical wind speeds for the Liujiaxia Bridge with considering the central stabilizer measure at different wind attack angles.

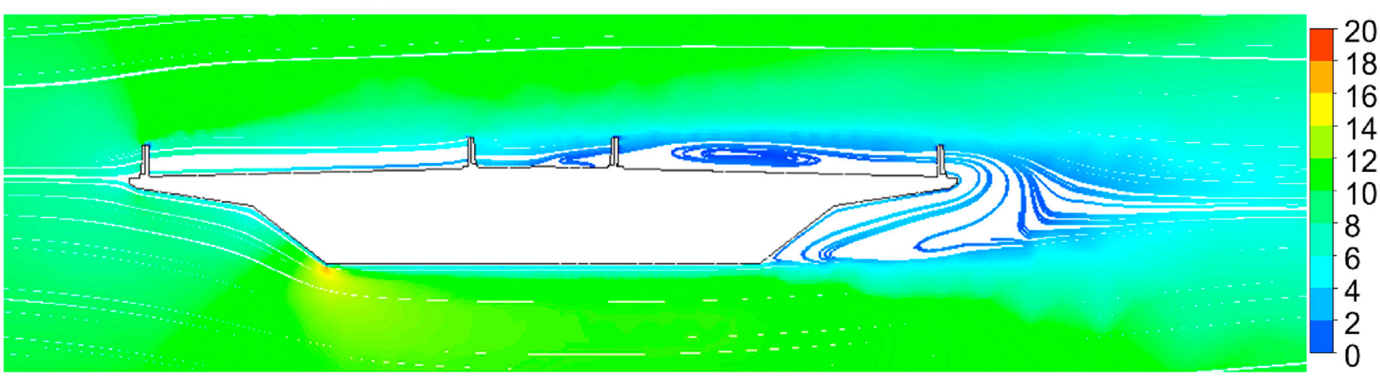
In Case 1, a downward central stabilizer with 1.12 m in height is installed at the lower side of the deck, the height of which is 0.28 times



(a) Deck model with the upward central stabilizer



(b) Deck model with sealed traffic barrier



(c) Deck model with partially sealed (side and central) traffic barrier

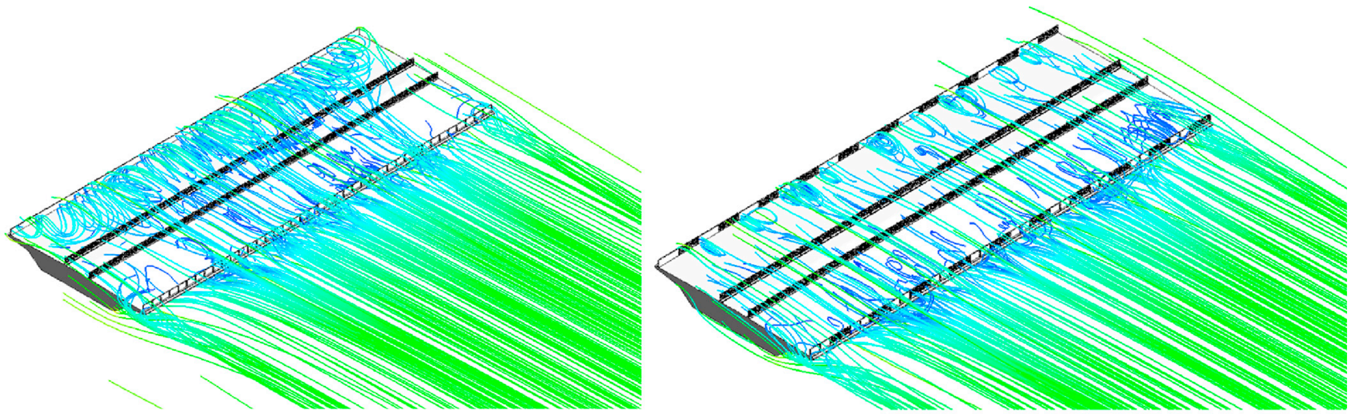
Fig. 19. Flow field around box girder section with different mitigation configurations.

the height of the steel truss girder. Compared with the original scheme, the flutter critical wind speed was significantly increased from 48.2 m/s to more than 70.4 m/s at wind attack angle -3° The flutter critical wind speed at wind attack angle 0° is also increased slightly from 43.8 m/s to 50.6 m/s. However, the flutter critical wind speed at wind attack angle +3° decreased slightly from 41.6 m/s to 37.4 m/s. Therefore, the downward central stabilizer can improve the flutter stability of the steel truss bridge at negative and 0° wind attack angles. When the wind attack angle of is positive, the flutter stability of the steel truss bridge is reduced. As a result, it is necessary to find other ways to improve the flutter stability of the bridge at the positive wind attack angle.

In Cases 2 to 4, the size of the downward central stabilizer remains unchanged, whereas the height of the upward central stabilizer is 0.8 m, 1.12 m, and 1.36 m, respectively. In Case 4, the upward central stabilizer is not continuously arranged on the bridge deck (15.6 m in width); the reason for this setting is that the continuously installed upward central stabilizer may cause traffic jams once a traffic accident occurs on the bridge deck since the vehicles may not be able to change lanes or turn around on the bridge. Therefore, the upward central stabilizer is installed in the form of segments with the length of each stabilizer and interval between two stabilizers being 30 m.

With the presence of both the upper and downward central stabilizers on the deck, the flutter critical wind speed is increased at each wind attack angle. Among Cases 1 to 3, by keeping the height of the downward central stabilizer unchanged, the flutter stability of each wind attack angle increases as the height of the upward central stabilizer increases. It is noted that Case 3 has the best flutter stability. However, in Case 3, the upward central stabilizer is continuously installed and this may cause traffic jams and other issues. Therefore, it is not advocated in practice. Compared with Case 3, the flutter critical wind speeds in Case 4 at wind attack angles 3° and 5° drop sharply. Specifically, for the wind attack angle 5°, the flutter critical wind speed is only 39.6 m/s, which cannot meet the requirements of the Chinese wind-resistant design specification.

In summary, the downward central stabilizer can result in an increase



(a) Deck model with sealed traffic barrier (b) Deck model with partially sealed traffic barrier

Fig. 20. Flow field around the bridge section models with sealed and partially sealed traffic barriers by 3D numerical simulations.

on the flutter critical wind speed at the negative wind attack angles. However, using both the upper and downward central stabilizers at the same time is better than using the downward central stabilizer alone. Increasing the height of the upward central stabilizer can result in an increase on the flutter critical wind speed. When the upward central stabilizer is arranged in segments with intervals, the flutter critical wind speed at the positive wind attack angle is dramatically reduced. Therefore, taking into account the actual situation for the Liujiaxia Bridge, the central stabilizer cannot effectively increase the flutter critical wind speed at each wind attack angle and other aerodynamic measures are needed.

To facilitate the aerodynamic analysis of the flutter divergence mechanism, Fig. 23 shows the key flutter derivatives, versus the reduced frequency *U/fB*, in cases with variants of the central stabilizer measure for the Liujiaxia Bridge based on the wind tunnel tests. For the key derivatives, A_2^* is the direct derivative (dominant one), representing the torsional aerodynamic damping effect on the moment coefficient, and its positive value refers to negative damping and vice versa. The coupling derivative (sub-derivative) H_2^* reflects the aerodynamic lift induced by the rotation speed (Yang et al., 2015). The flutter derivatives in Case 4 were not measured since its flutter critical wind speed at wind attack angle 5° does not surpass the threshold and therefore, they are not shown in this figure. It is noted in Fig. 23 that: (1) The values of the derivative A_2^* (key parameter) associated with Case 1 and the original scheme change from negative to positive with the increase of the reduced frequency *U/fB*, indicating the formation of negative aerodynamic damping. However, since the downward central stabilizer is set in Case 1, the corresponding reduced wind speed is relatively higher than that associated with the original scheme. Meanwhile, the values of the flutter derivative A_2^* in Cases 2 and 3 are negative and become larger in the absolute value along with the increase of *U/fB*, resulting in larger positive aerodynamic damping ratios which contribute to the aerodynamic stability of the structure. As the value of A_2^* in Case 3 is larger than that in Case 2 for most of the time, the bridge flutter stability associated with Case 3 is deemed as the most favorable one; (2) Another dominant derivative A_3^*imposes a small impact on the flutter critical wind speed for the truss-girdered suspension bridge, and its values for different cases are close to each other and they follow similar pattern with the increase of *U/fB*; and (3) The value of the coupling derivative H_2^* is rather larger than that in other cases and this is beneficial to the flutter stability of the truss-girdered suspension bridge. The above analysis highlights that different mitigation measures should modify the aerodynamic shape of the bridge in certain ways, thus resulting in changes for the aerodynamic coupling effect under the self-excited force.

4.4. Guide plate measure for Liujiaxia Bridge

At this step, the upward central stabilizer was removed while keeping the downward one which is fixed at 1.12 m (except Case 8). Here five variants of the guide plate measure are considered for the Liujiaxia Bridge, named Cases 5 to 9, and their settings are given in Table 9. The width of the guide plate is 1.28 m (except Case 6), 0.32 times the height of the truss girder beam. In Cases 5 to 9, the position and inclination of the guide plate on the steel truss girder were changed and their influence, combined with the downward central stabilizer, on the flutter stability of the steel truss suspension bridge was investigated. Table 10 shows the flutter critical wind speed for the Liujiaxia Bridge with considering the guide plate measure at different wind attack angles.

Regarding Case 5, it can be seen that the combination of the upward and downward central stabilizers performs better on the improvement of the flutter stability of the steel truss bridge than the combination of the downward central stabilizer and guide plate. Specifically, with the presence of the horizontal guide plate in Case 5, the flutter critical wind speed at wind attack angle 3° is largely reduced, only 17.6 m/s. In Case 6, the width of the horizontal guide plate is extended from 1.28 m to 2.0 m. As a result, the flutter critical wind speed at wind attack angle 3° is raised from 17.6 m/s to 41.6 m/s. However, the flutter critical wind speed at wind attack angle 0° was reduced from 66 m/s to 43.8 m/s. It can be seen that when the width of the horizontal guide plate is small ($L_1 = 0.32H$, i.e., 1.28 m), the flutter stability performance at wind attack angle 3° cannot be improved. Increasing the width of the horizontal guide plate ($L_1 = 0.5H$, i.e., 2.0 m), the flutter stability at wind attack angle 3° can be improved, but it is reduced at wind attack angle 0°. The wider horizontal guide plate ($L_1 = 0.5H$) enhances the balance of flutter stability of the steel truss bridge at different attack angles and therefore, there is no extremely poor flutter stability at any wind attack angles.

In Case 7, the guide plate, with the same width as in Case 5, is installed with tilt angle 30°. The flutter critical wind speed at wind attack angle 3° increased from 17.6 m/s to 39.6 m/s. It can be seen that when the width of the guide plate is the same, the inclined installation is better than the horizontal installation. It is noted in Cases 8 and 9 that the guide plate installed inside the steel truss bridge are not as effective as that installed outside of the bridge.

In summary, the vibration suppression effect of the guide plate measure is not as good as that of the central stabilizer measure. The guide plate, installed outside of the steel truss bridge, horizontally or with a certain inclination angle, performs better than that installed inside the steel truss bridge in terms of the vibration suppression effect. With the same width of the guide plate, the inclined installation outside of the steel truss bridge performs better than the horizontal installation.

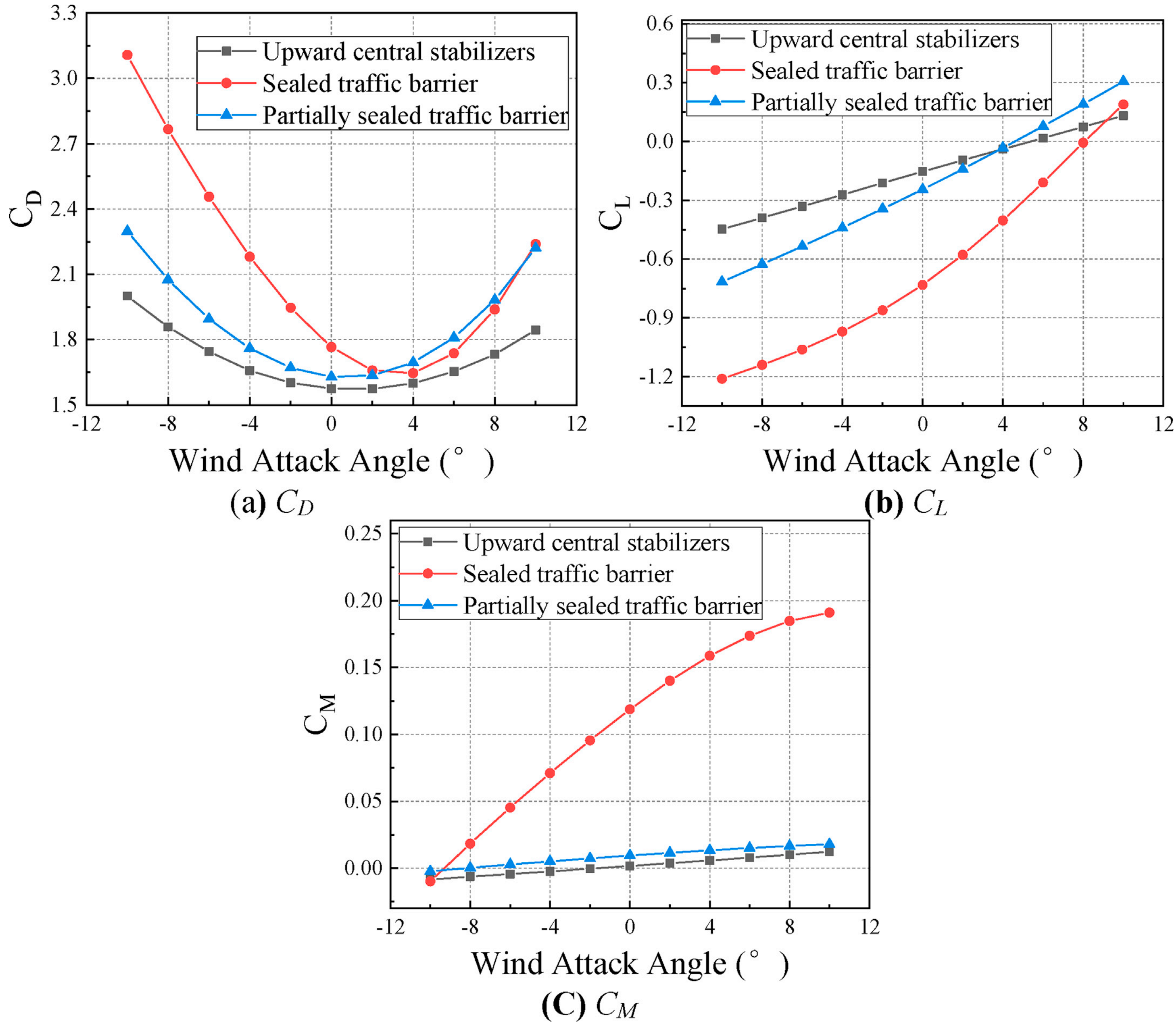


Fig. 21. Aerodynamic coefficients for the section model with three distinctive mitigation measures.

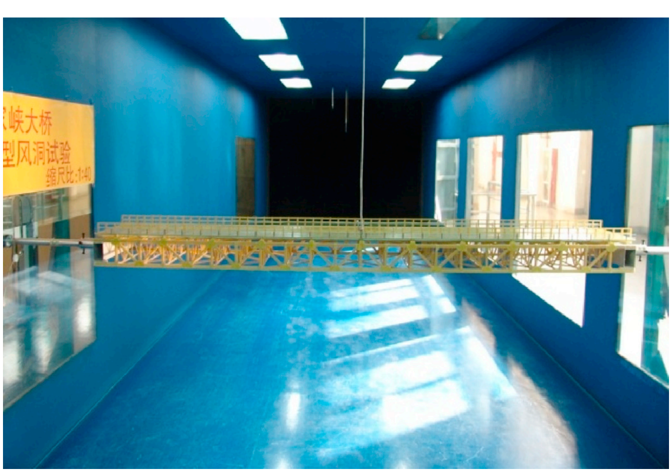


Fig. 22. Section model of Liujiaxia Bridge in wind tunnel.

Table 4
Parameters for the section model of the Liujiaxia Bridge in the flutter test.

Properties	Parameter	Unit	Prototype	Scale ratio	Section model
width	B	M	15.6	1/40	0.39
Mass per unit length m	M	kg/m	12800	$1/40^2$	8.00
Mass moment of inertia per unit length	J_m	$kg.m^2/m$	557000	$1/40^4$	0.218
Vertical frequency	f_b	Hz	0.196	40/4.38	1.79
Torsional frequency	f_t	Hz	0.453	40/4.38	4.14
Vertical damping ratio	ξ_b	%	0.5	1	0.399
Torsional damping ratio	ξ_t	%	0.5	1	0.364

Table 5
Test conditions for the section model of the Liujiaxia Bridge.

Section model (Scale ratio)	Wind attack angle (°)	Wind speed ratio	Flow field
1:40	−5, −3, 0, +3, +5	1:4.38	Uniform oncoming flow

Table 6
Flutter critical wind speed for the Liujiaxia Bridge in the uniform flow field.

Wind attack angle (°)	Flutter critical wind speed (m/s)	
	section model	real bridge
−3	11.0	48.2
0	10.0	43.8
+3	9.5	41.6

Table 7
Variants of the central stabilizer measure for the Liujiaxia Bridge.

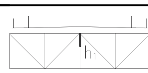
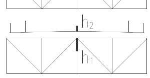
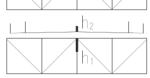
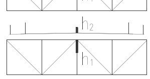
Case	Description	Central stabilizer	
		Schematic diagram	Parameter
1	Downward central stabilizer		$h_1 = 1.12$ m
2	Upward central stabilizerI+ downward central stabilizer		$h_1 = 1.12$ m, $h_2 = 0.8$ m
3	Upward central stabilizerII+ downward central stabilizer		$h_1 = 1.12$ m, $h_2 = 1.12$ m
4	Upward central stabilizer III+ downward central stabilizer (the length of each stabilizer segment is 30 m with the interval 30 m)		$h_1 = 1.12$ m, $h_2 = 1.36$ m

Table 8
Flutter critical wind speed for the Liujiaxia Bridge with the central stabilizer measure.

Case	Flutter critical wind speed (m/s)				
	-5°	-3°	0°	3°	5°
1		>70.4	50.6	37.4	
2	>70.4	>70.4	>70.4	>70.4	48.2
3	>70.4	>70.4	>70.4	>70.4	>70.4
4	>70.4	>70.4	>70.4	>68.2	39.6

Increasing the width of the horizontal guide plate can improve the flutter stability at wind attack angle 3°, but the increase is limited. Therefore, the guide plate measure did not significantly improve the flutter stability at the each wind attack angle for this steel truss suspension bridge, especially at wind attack angle 3°.

Since the Flutter critical wind speed at wind attack angle -3° is notably larger than the values at the other two attack angles, the corresponding key flutter derivative $A_2^*$is examined based on available data, as shown in Fig. 24. It is noted that the absolute values of the dominant derivative A_2^* in Cases 5 and 7 are generally larger than that in Cases 8 and 9, indicating that the flutter stability associated with Cases 5 and 7 is superior to that in the later cases. This confirms that the positive aerodynamic damping induced by the pneumatic self-excited force can suppress the flutter vibration of the truss-girdered suspension bridge and a larger aerodynamic damping brings more benefits to the bridge flutter stability.

4.5. Sealed side traffic barrier measure for Liujiaxia Bridge

Since both the downward central stabilizer and guide plate measures cannot desirably improve the flutter stability for the Liujiaxia Bridge at different wind attack angles, the measure for sealing the traffic barrier on both sides of the bridge deck were investigated, i.e., Case 10. In Case 10, the height of the traffic barrier is set as 2.12 m, where the upper part with height 1.04 m is completely sealed and the lower part with height 1.08 m adopts an arrangement with regular intervals, as shown in Fig. 25. The ventilation rate for Case 10 is 35%. It should be noted that in Case 10, the downward central stabilizer with height 1.12 m and the horizontal guide plate with width 2.0 m, as considered in Case 6, is also presented.

Table 11 shows the flutter critical wind speed for the Liujiaxia Bridge with the sealed side traffic barrier measure. It is worth noting that sealing the side traffic barrier, though with the upward central stabilizer cancelled, can play similar effect to the upward central stabilizer such that the flutter critical wind speed at different wind attack angles is improved. In Case10, favorable flutter stability at different wind attack angles have achieved, thus it can meet the requirements of the Chinese wind-resistant design specification.

Under the condition that the combined measures of the downward central stabilizer and guide plate cannot adequately improve the flutter stability for the Liujiaxia Bridge at each wind attack angle, the sealed side traffic barrier plays the role of the upward central stabilizer. As a result, the flutter stability for the Liujiaxia Bridge at each wind attack angle has been improved, meeting the code requirements. In addition, the sealed side traffic barrier does not cause VIV issues for the Bridge. The aerodynamic coefficients associated with Case 10 are compared with that associated with the original scheme, as shown in Fig. 26. It is observed in this figure that the combined measures in Case 10 pose relatively small impacts on the drag coefficient, resulting in a slight increase and decrease on the value at, respectively, negative and positive attack angles. For the moment coefficient, there is negligible difference between the two cases. However, the combined measures in Case 10 play a notable role for the lift coefficient, especially in scenarios at negative wind attack angles, where the lift coefficient is notably reduced, thereby benefiting the structural aerodynamic stability.

5. Improving both VIV and flutter performance for Chongqing Hechuan Bridge

5.1. Experimental set-up for Chongqing Hechuan Bridge

In the experiment for the Chongqing Hechuan Bridge, the scale of the section model was set as 1:35 with 1.5 m in length. Table 12 lists the parameters for the section model in the VIV and flutter test. The tested wind attack angles are -3°,0°and +3°, and the tested cases are listed in Table 13. Three acceleration sensors and displacement laser sensors were mounted to monitor the vibrations of models. The sampling frequencies of the acceleration and displacement sensors were set as 1000 Hz and 256 Hz, respectively, and the sampling duration for each case was 60 s.

5.2. Aerodynamic performance of Chongqing Hechuan Bridge

Fig. 27 shows the vertical and torsional VIV responses of the Chongqing Hechuan Bridge at wind attack angles -3°, 0°and +3°. Again, the maximum values of the responses and the wind speed have been converted to the prototype scale. The vertical VIVs were observed at wind attack angles -3°, 0°and +3°. As the wind attack angle changes from negative to positive, the amplitude of the vertical VIV increases gradually. The maximum vertical VIV amplitude is 0.171 m, which occurs at the wind attack angle +3°and the wind speed is 17.0 m/s. The torsional VIV were also observed at wind attack angles -3°, 0°and +3°.The maximum values of torsional VIV at 0°and +3°are 0.820°and 0.825°, respectively, and the corresponding wind speeds are 17.0 m/s and 32.3 m/s.

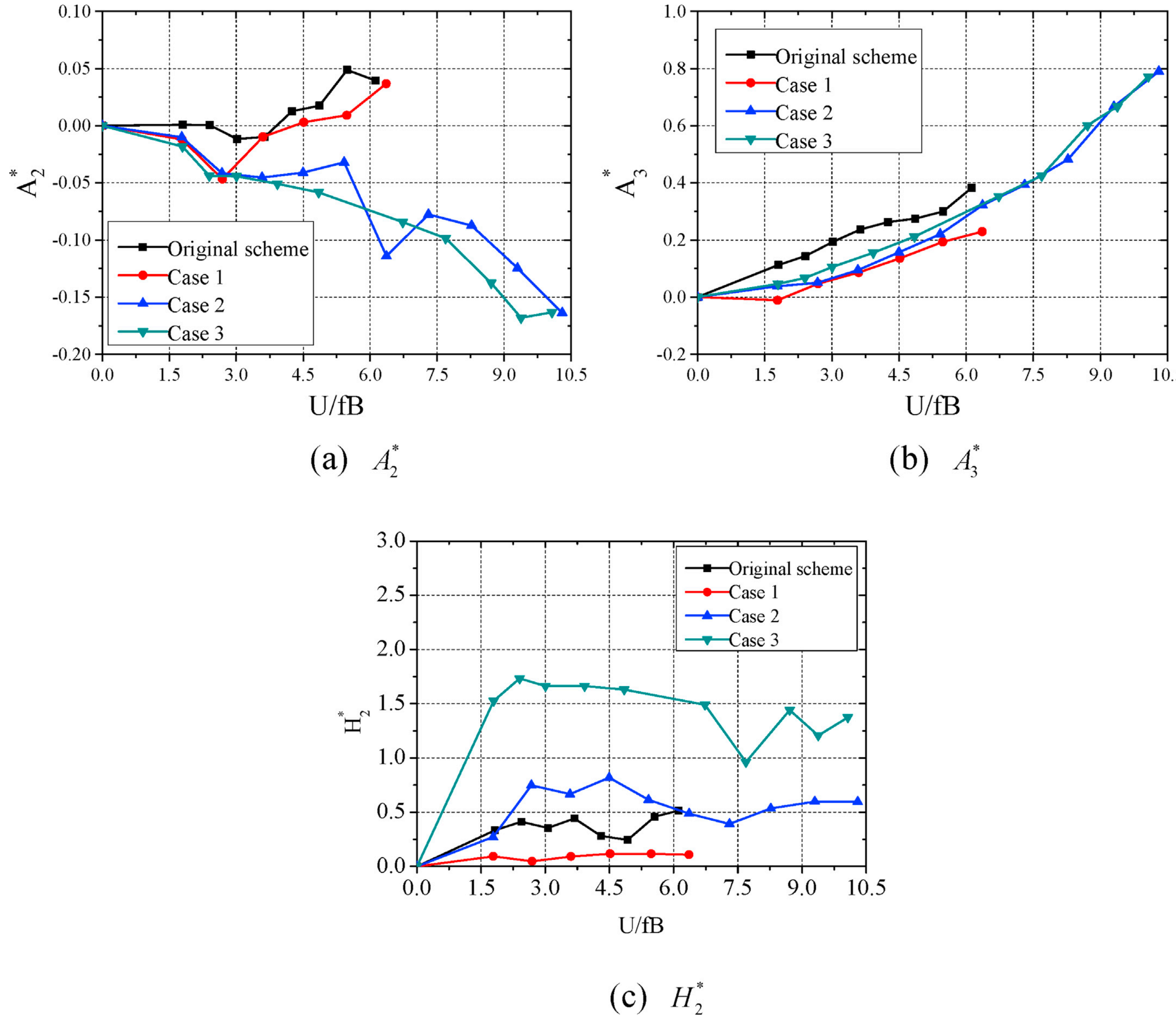


Fig. 23. Comparison of flutter derivatives in cases with variants of the central stabilizer measure for the Liujiaxia Bridge.

Table 9
Variants of the guide plate measure for the Liujiaxia Bridge.

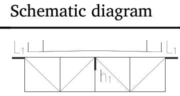
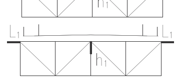
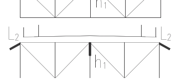
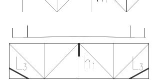
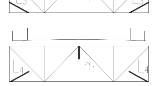
Case	Description	Guide plate: Schematic diagram	Guide plate: Parameter
5	Downward central stabilizer + horizontal guide plate		$h_1 = 1.12$ m, $L_1 = 1.28$ m
6	Downward central stabilizer + horizontal guide plate		$h_1 = 1.12$ m, $L_1 = 2.0$ m
7	Downward central stabilizer + tilting guide plate		$h_1 = 1.12$ m, $L_2 = 1.28$ m (30°tilt angle)
8	Downward central stabilizer + build-in tilting guide plate		$h_1 = 1.28$ m, $L_3 = 1.28$ m (−30°tilt angle)
9	Downward central stabilizer + build-in tilting guide plate		$h_1 = 1.12$ m, $L_4 = 1.28$ m (30°tilt angle)

Table 10
Flutter critical wind speed for the Liujiaxia Bridge with the guide plate measure.

Case	Flutter critical wind speed (m/s)		
	-3°	0°	3°
5	>70.4	66.0	17.6
6	>66.0	43.8	41.6
7	>70.4	66.0	39.6
8	66.0	48.2	41.8
9	57.2	39.6	33.0

Based on the wind-resistant design specification for highway bridges (Ministry of Communications of the People's Republic of China, 2004), the amplitude limitation of vertical and torsional VIVs ($[h_\alpha]$and $[\theta_\alpha]$) are respectively defined as follows:

$$[h_\alpha] = 0.04/f_h = 0.130m \tag{4}$$

$$[\theta_\alpha] = 4.56/(Bf_t) = 0.380^\circ \tag{5}$$

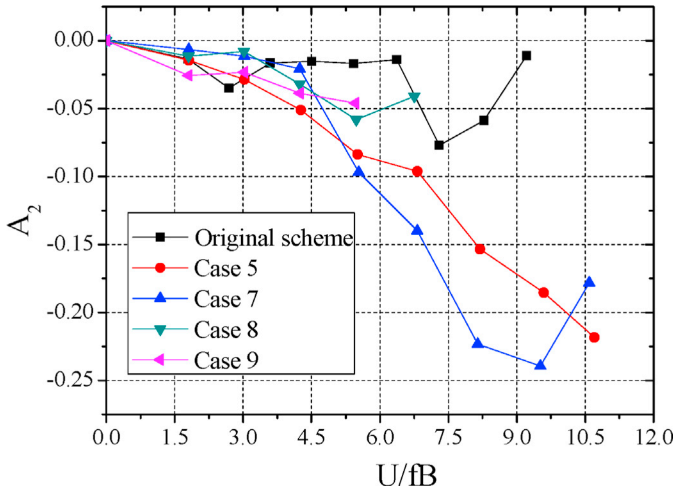


Fig. 24. Comparison of the flutter derivative A_2^* in cases with variants of the guide plate measure for the Liujiaxia Bridge.

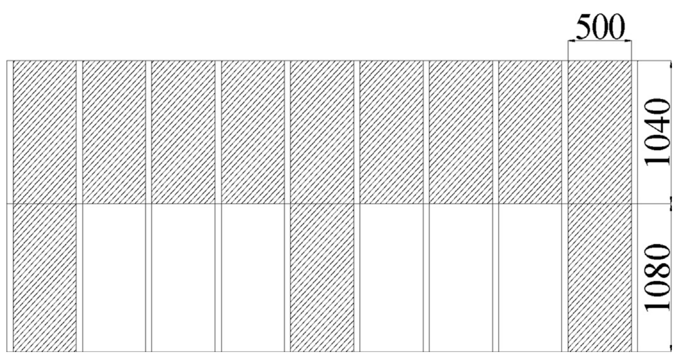


Fig. 25. Sealed side traffic barrier measure for the Liujiaxia Bridge (Case 10) (Unit: mm).

Table 11
Flutter critical wind speed for the Liujiaxia Bridge with the sealed side traffic barrier measure.

Case	Flutter critical wind speed (m/s)				
	-5°	-3°	0°	3°	5°
10	66.0	>70.4	66.0	>70.4	61.6

where f_h, f_t, and B are the vertical frequency, torsional frequency, and width of the Chongqing Hechuan Bridge (see Table 12). Compared with the VIV amplitude limitations in the Chinese specification, the maximum amplitudes of the vertical VIV at wind attack angle +3° and the torsional VIVs at wind attack angles 0°and +3° cannot satisfy the design requirements. It is noted that there are two lock-in districts for the vertical and torsional VIVs, and the wind speed of the first lock-in district is approximately from 12 m/s to 18 m/s. In addition, due to the bridge site is near to the mountain, the flow conditions are uncertain and complicated, and large wind attack angles exist occasionally. Considering the complicated surrounding terrains at the bridge site and that the vertical and torsional VIV's intensity occurred at such low wind speed, efforts should be made to improve the VIV performance of the Chongqing Hechuan Bridge.

Fig. 28 shows the vertical and torsional damping ratio curves versus the wind speed for the Chongqing Hechuan Bridge at wind attack angles -3°, 0°, and +3°, where the wind speed is the test wind speed in the wind tunnel. It is observed that the vertical damping ratio increases as the test

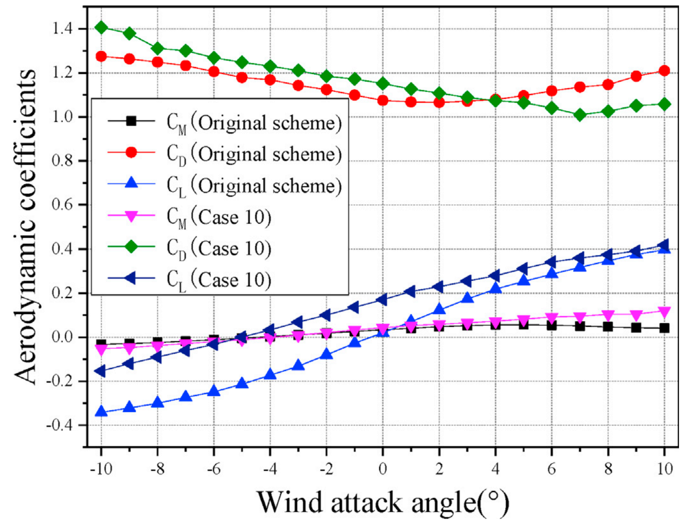


Fig. 26. Aerodynamic coefficients associated with Case 10 for the Liujiaxia Bridge.

wind speed increases. The torsional damping ratio increases first and then decreases with the increase of the test wind speed at wind attack angles 0° and +3°. When the test wind speed, with respect to the wind attach angles 0°and +3°, is 7.1 m/s and 7.3 m/s, respectively, the torsional damping ratio is close to zero, and the torsional flutter occurs. Table 14 shows the critical wind speed of the flutter at the different wind attack angles for the Chongqing Hechuan Bridge in the uniform flow field. By converting the test wind speed to the prototype values, the flutter critical wind speeds of the Hechuan Bridge at wind attack angles 0° and +3° are 34.8 m/s and 35.8 m/s, respectively.

Based on the wind-resistant design specification for highway bridges (Ministry of Communications of the People's Republic of China, 2004), the flutter checking wind speed ($[V_{cr}]$) is defined as follows:

$$[V_{cr}] = 1.2\mu_f V_d = 50.6m/s \tag{6}$$

where $\mu_f = 1.29$ is the fluctuation wind speed correction factor and $V_d = 32.7$ m/s is the design standard wind speed of the Chongqing Hechuan Bridge.

It can be seen from the wind tunnel test results that the flutter stability of the Hechuan Bridge does not meet the requirements of the Chinese wind-resistant design specification at wind attack angles 0° and +3°. Therefore, measures need to be taken to improve the aerodynamic stability of the Chongqing Hechuan Bridge, including VIV stability at wind attack angles -3°, 0°and +3°and flutter stability at wind attack angles 0°and +3°.

5.3. Aerodynamic measures for Chongqing Hechuan Bridge

Considering the characteristics of the slotted box girder and the operability of the optimized design, in order to improve the aerodynamic stability of the Chongqing Hechuan Bridge, two aerodynamic measures, the fairing and inspection vehicle rail, were firstly investigated in the wind tunnel test. The schematic diagrams of these two aerodynamic measures are shown in Figs. 29 and 30. For the slotted box girder with two steel boxes, a sidewalk is maintained on one steel box. The high sidewalk pedestal is more likely to cause airflow separation and vortex shedding. The wind tunnel test shows that the aerodynamic stability of the bridge is more unfavorable when the sidewalk is on the upstream side of the incoming flow. Therefore, the fairing at the sidewalk side is enlarged such that it can wrap the sidewalk pedestal, as shown in Fig. 31.

Table 15 presents the VIV and flutter wind tunnel test results associated with the faring and vehicle inspection rail measures, in total three

Table 12
Parameters for the section model of the Chognqing Hechuan Bridge in the VIV and flutter test.

Content	Properties	Parameter	unit	Prototype	Scale ratio	Section model
Basic parameters	Width	B	m	19	1/35	0.543
	Mass per unit length m	M	kg/m	13000	$1/35^2$	10.612
	Mass moment of inertia per unit length m	J_m	$kg.m^2/m$	1330000	$1/35^4$	0.469
VIV test	Vertical frequency	f_b	Hz	0.307	35/3.4	3.160
	Torsional frequency	f_t	Hz	0.631	35/3.4	6.496
	Vertical damping ratio	ξ_b	%	0.5	1	0.424
	Torsional damping ratio	ξ_t	%	0.5	1	0.397
Flutter test	Vertical frequency	f_b	Hz	0.307	35/4.9	2.193
	Torsional frequency	f_t	Hz	0.631	35/4.9	4.507
	Vertical damping ratio	ξ_b	%	0.5	1	0.438
	Torsional damping ratio	ξ_t	%	0.5	1	0.405

Table 13
Test conditions for the section model of the Chongqing Hechuan Bridge.

Content	wind attack angle (°)	wind speed ratio	Flow field
VIV	−3°, 0°, +3°	1:3.4	Uniform oncoming flow
Flutter	−3°, 0°, +3°	1:4.9	Uniform oncoming flow

cases, where symbol '—' indicates that this test has not been carried out. It can be seen in these cases that obvious VIV at wind attack angle 3° is still observed, failing to meet the requirements of the Chinese wind-resistant design specification. In order to modify the original scheme as little as possible, the measures of sealing the central traffic barrier, side traffic barrier, and sidewalk traffic barrier are considered, as shown in Figs. 32–34. Fig. 35 shows the test photograph of the side traffic barrier and sidewalk traffic barrier.

Table 16 shows the VIV and flutter wind tunnel test results associated with the sealed central traffic barrier, sealed side traffic barrier, and sealed sidewalk traffic barrier measures. It can be seen that in Case IV, the fully sealed central traffic barrier measure can increase the flutter critical wind speed of the bridge, but the improvement is limited; in addition, he VIV can still be observed in this case. As follows, to improve both the flutter and VIV stability of the bridge, two or more aerodynamic combination measures are required. In Cases V and VI, which are the combination measure of the fairing and fully sealed central traffic barrier and the combination measure of the fairing and inspection vehicle rail, respectively, the VIV of the bridge is not eliminated. In order to suppress the periodic vortex shedding, Case VII adopted a combination measure of the sealed side traffic barrier with intervals and sealed sidewalk traffic barrier with intervals. As a result, the flutter critical wind speed is improved to some extent, but the VIV is still observed. In Case VIII, the fairing measure is added on the basis of Case VII, and therefore, the flutter critical wind speed is further improved, which can meet the requirements of Chinese wind-resistant design specification. However, the VIV still exists in this case. Based on Case VIII, Case IX covers the fully sealed central traffic barrier and inspection vehicle rail. Under the action of these five combined measures, the VIV at wind attack angles -3°, 0° and +3° is eliminated and the flutter critical wind speeds at the three wind attack angles are all larger than 63.5 m/s, thus meeting the code requirements. As such, the fully sealed central traffic barrier, along with the sealed sidewalk traffic barrier and sidewalk traffic barrier with intervals, can effectively modify the vortex shedding form and frequency of the bridge and suppress the VIV of the bridge while improving the flutter critical wind speed. Fig. 36 shows the VIV test results of the Chongqing Hechuan Bridge under the combination aerodynamic measures in Case IX.

In order to obtain better aerodynamic stability, the height of the sidewalk base was reduced from 0.34 m to 0.14 m, as exhibited in Case 10. In this case, the side traffic barrier and sidewalk traffic barrier are not sealed, a shorter inspection vehicle rail B-② is used, aiming to make the bridge more streamlined and eliminate the VIV. As a result, the flutter critical wind speed of the bridge is 56.3 m/s at wind attack angle +3°, which can meet the requirements of Chinese wind-resistant design specification. The flutter critical wind speeds at wind attack angles 0° and +3° are much larger than the flutter checking wind speed. It can be seen

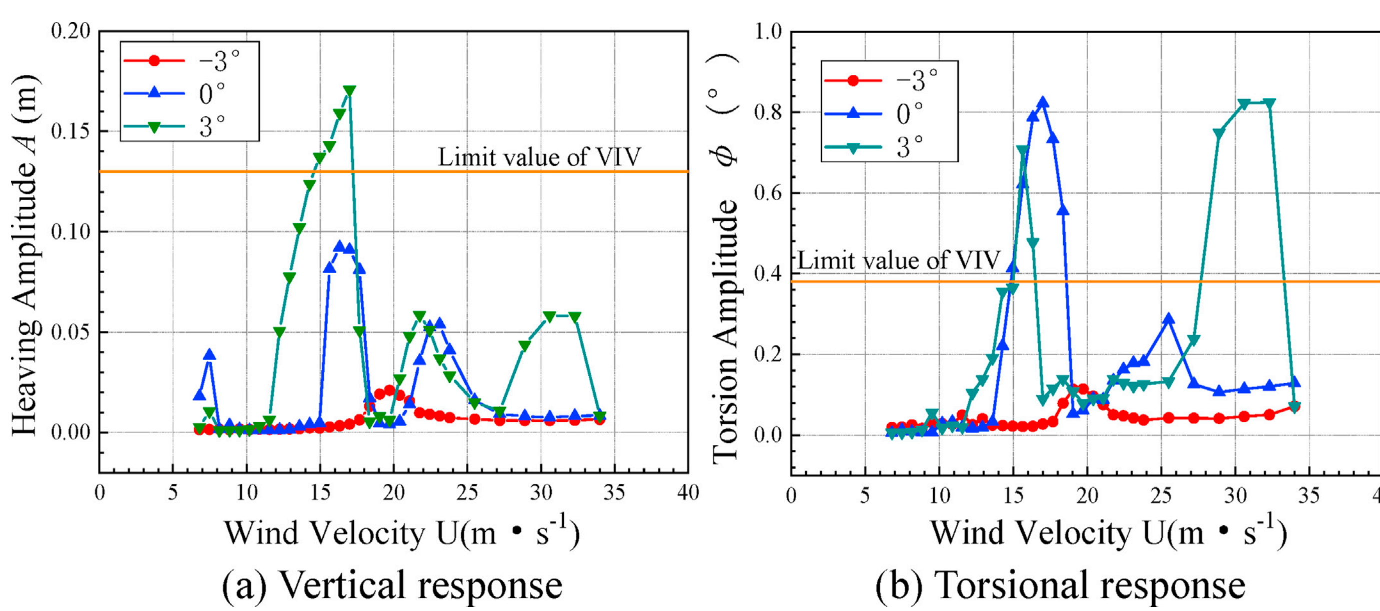


Fig. 27. Vertical and torsional VIV responses of Chongqing Hechuan Bridge.

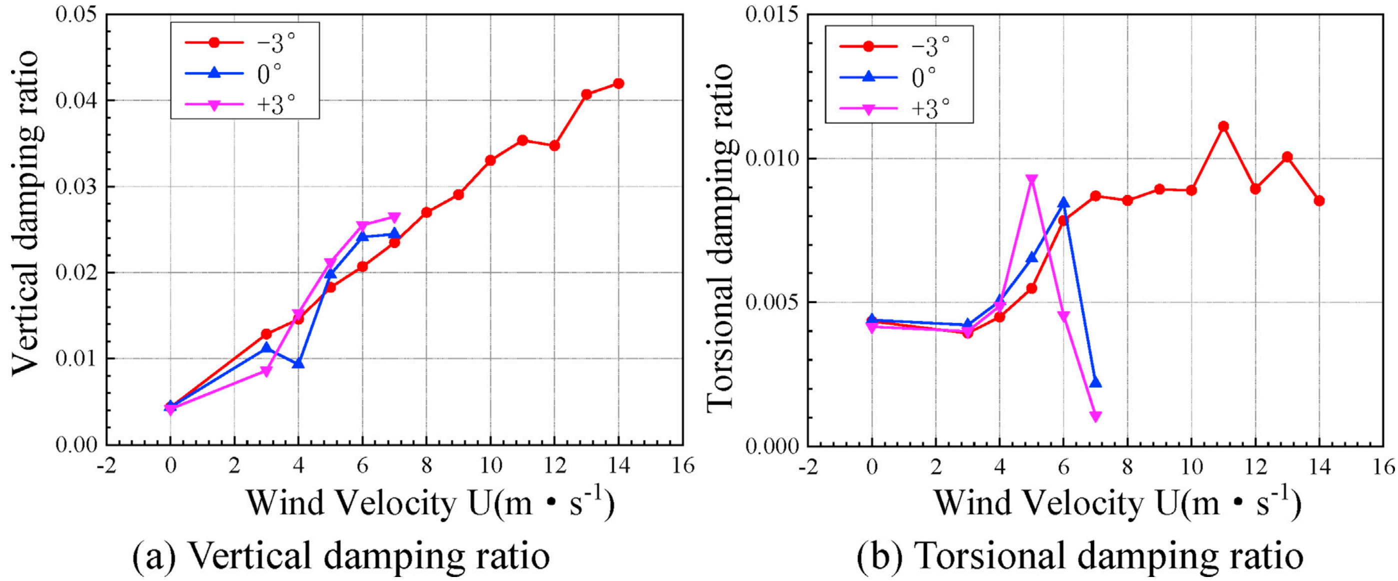


Fig. 28. Vertical and torsional damping ratios for the Chongqing Hechuan Bridge.

Table 14
Flutter critical wind speed for the Chongqing Hechuan Bridge.

Wind attack angle (°)	Flutter critical wind speed (m/s)	
	section model	real bridge
−3	>14.0	>68.6
0	7.1	34.8
+3	7.3	35.8

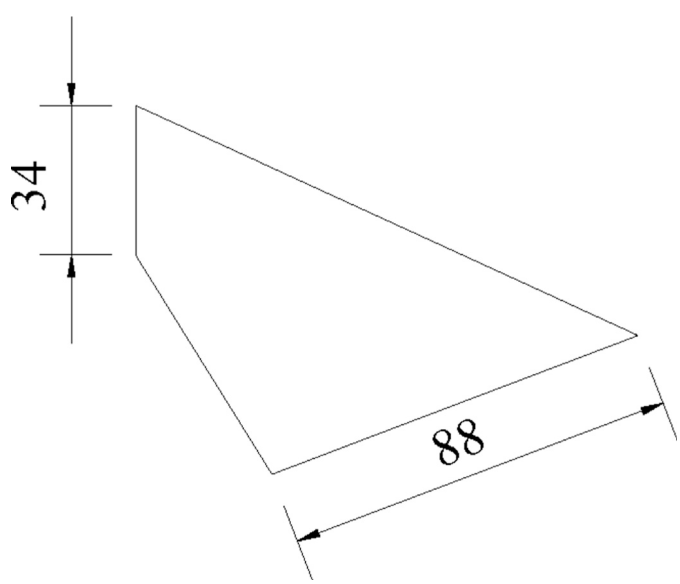


Fig. 29. Schematic diagram of the fairing measure (A) (Unit: cm).

that reducing the height of the sidewalk base and the height of the inspection vehicle rail will result in a better streamlined form of the bridge, thus improving the aerodynamic stability of the bridge.

Compared among the aforementioned cases, the combination measures in Case Ⅸ have the best vibration suppression effect. It shows that the sealed side traffic barrier and sidewalk traffic barrier in the form of D-② and E−② can effectively suppress the VIV and increase the flutter critical wind speed.

6. Effect of sealing form for traffic barrier on VIV and flutter performance

In previous sections, desirable aerodynamic measure or combination measures regarding the sealed traffic barrier for suppressing VIV and increasing the flutter critical wind speed in three distinctive bridge projects have been attained. This section explores the efficacy of additional variants of the sealing form for the traffic barrier on the VIV and flutter performance.

6.1. *Effect of Vertical and Horizontal seals on VIV*

This section extends the research on the aerodynamic measures for the Hongkong-Zhuhai-Macao Bridge. Here, Measure 3 with the horizontally sealed traffic barrier and ventilation rate 33% is proposed, as shown in Fig. 37.

Fig. 38 shows the vertical and torsional responses of the 1:50 section model of the Hongkong-Zhuhai-Macao Bridge with the aerodynamic Measure 3 at wind attack angles -5°, -3°, 0°, +3°and +5°. It can be seen that the vibration suppression effect associated with Measure 3 on the torsional VIV is better than that on the vertical VIV. The torsional VIVs of the original section were almost eliminated by the action of Measures 3. The maximum amplitude value of the vertical VIV at each wind attack

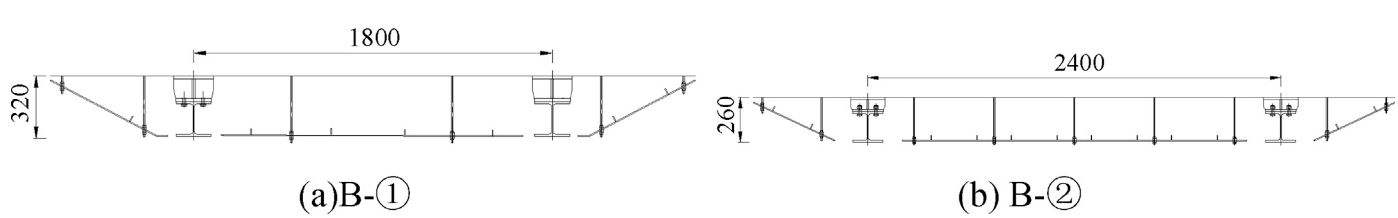


Fig. 30. Schematic diagram of the vehicle inspection rail measure (Unit: mm).

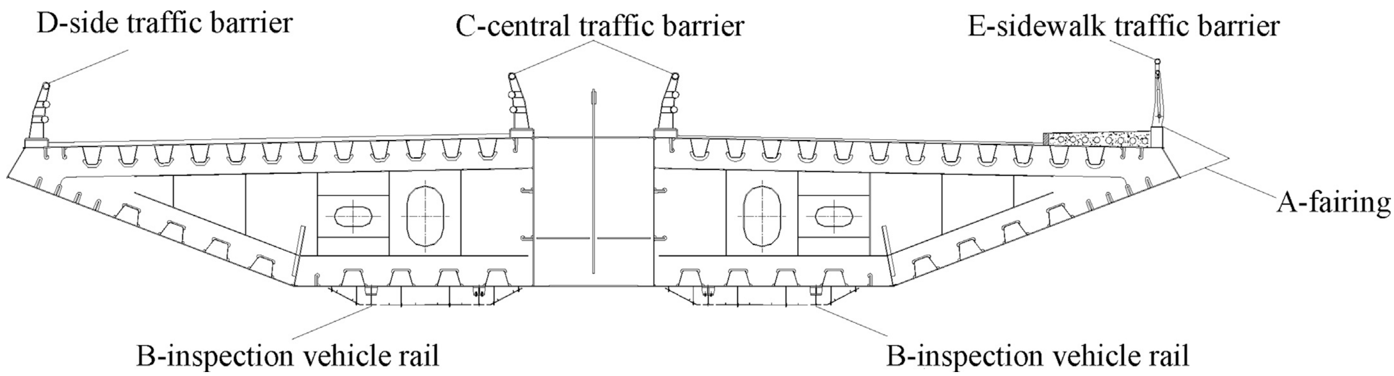


Fig. 31. Schematic diagram of aerodynamic measures, the fairing and inspection vehicle rail.

Table 15
VIV and flutter results associated with faring and inspection vehicle rail measures through the wind tunnel test.

Case	Aerodynamic measure	wind attack angle					
		+3°		0°		-3°	
		VIV	Flutter	VIV	Flutter	VIV	Flutter
I	A	observed	–	–	–	–	–
II	B-①	observed	–	–	–	–	–
III	B-②	observed	–	–	–	–	–

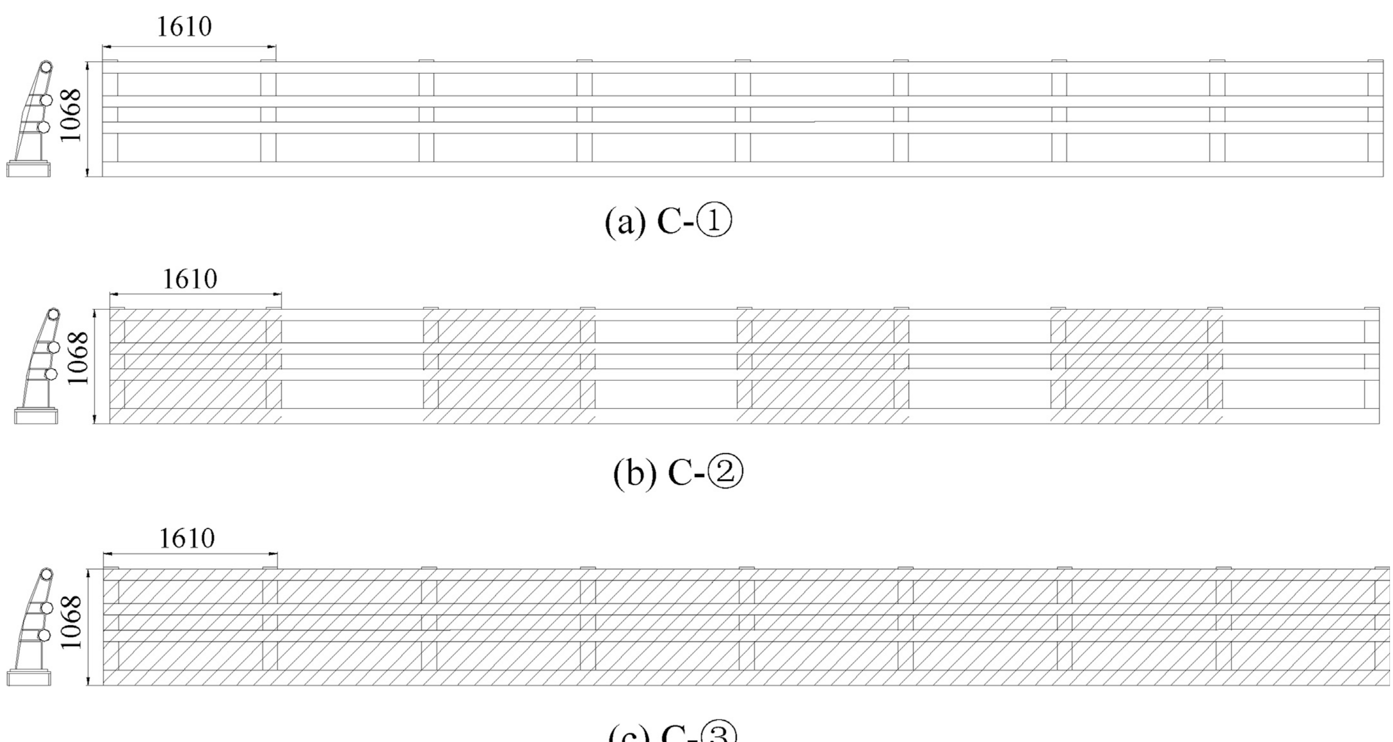


Fig. 32. Schematic diagram of the sealed central traffic barrier measure (Unit: mm).

angle was also dramatically reduced. The small amplitude value for the vertical VIV of the original section at wind attack angles 0° and +3° have disappeared. A large amplitude value for the vertical VIV can be observed in the wind speed range from 20 m/s to 26 m/s at wind attack angle +5°; however, this amplitude value does not exceed the limit value in the specification.

The comparisons between the wind tunnel results associated with Measures 2 and 3 (shown in Figs. 13 and 42, respectively) indicate that the vibration suppression effect associated with the horizontally sealed traffic barrier, i.e., Measure 3, on the vertical VIV is not superior to that associated with regular vertical intervals. However, Measure 3 performs slightly better on suppressing the torsional VIV than Measure 2.

6.2. *Effect of regular and irregular sealed traffic barriers on flutter*

This section conducts more research on the aerodynamic measures for

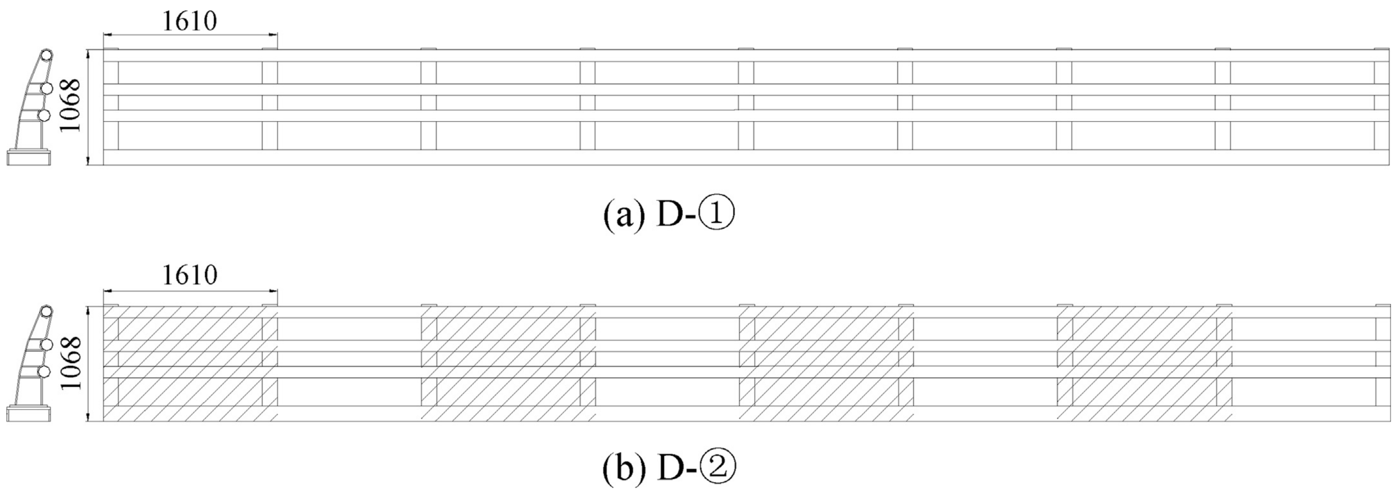


Fig. 33. Schematic diagram of the sealed side traffic barrier measure (Unit: mm).

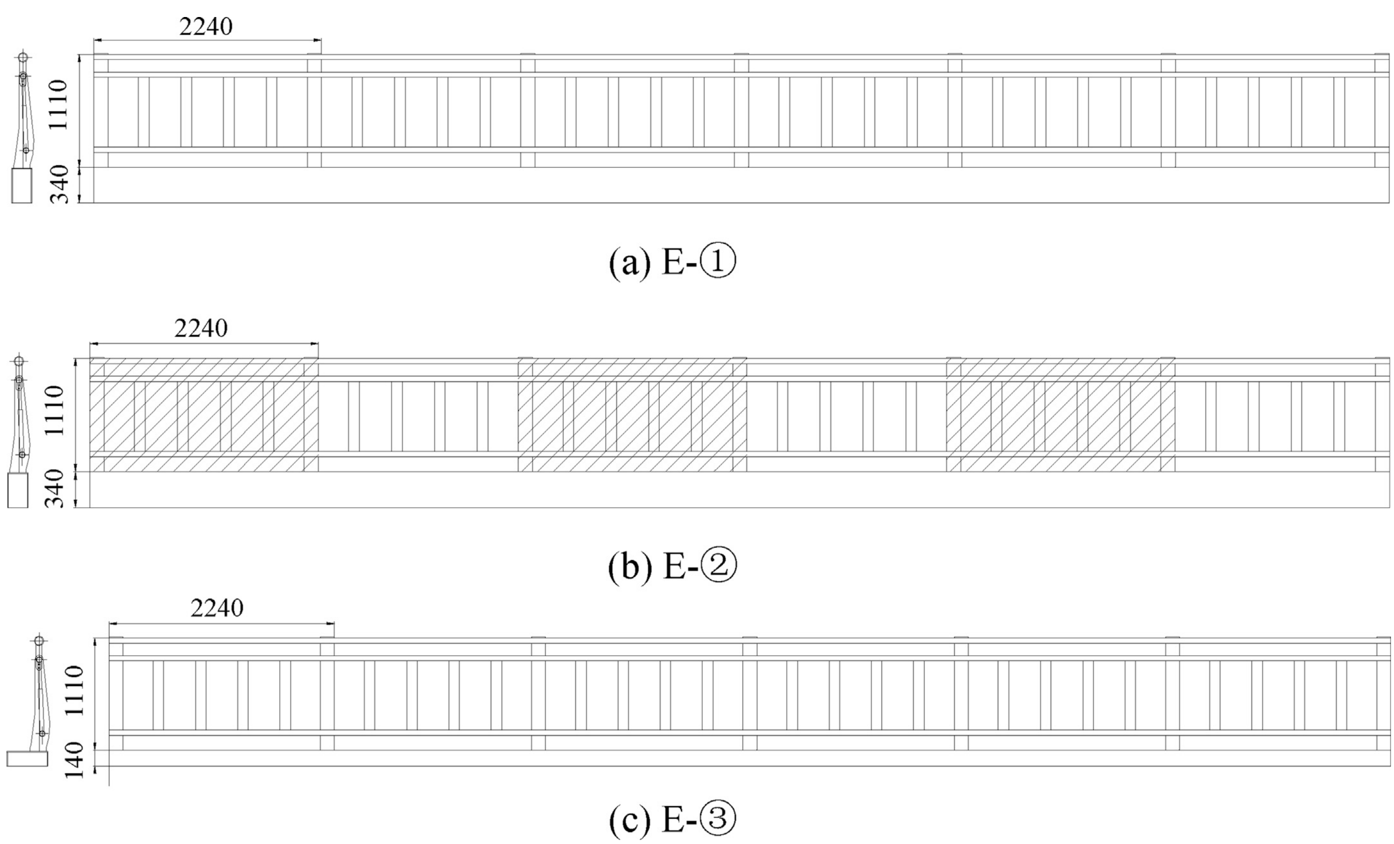


Fig. 34. Schematic diagram of the sealed sidewalk traffic barrier measure (Unit: mm). E−③ is similar to E−①, but the sidewalk base is reduced to 0.14 m.

the Liujiaxia Bridge. Here, two variants for sealing the traffic barrier on both sides of the Liujiaxia Bridge deck, Cases 11 and 12, were investigated, as shown in Fig. 39. In Case 11, the height of the traffic barrier is the same as that in case 10, and the upper part with height 1.04 m and the lower part with height 1.08 m adopt irregular arrangements, repeated along the bridge, as shown in Fig. 39(a). The width of each sealed block is 0.5 m, and the ventilation rate for case 11 is 60%. In Case 12, the height of the traffic barrier is shortened to 1.5 m, where the upper part with height 0.6 m is fully sealed and the lower part with height 0.9 m adopts a regular repeated arrangement. The ventilation rate for case 12 is 45%. For these two cases, the 1.12 m downward central stabilizer and the 2 m horizontal guide plate are also considered.

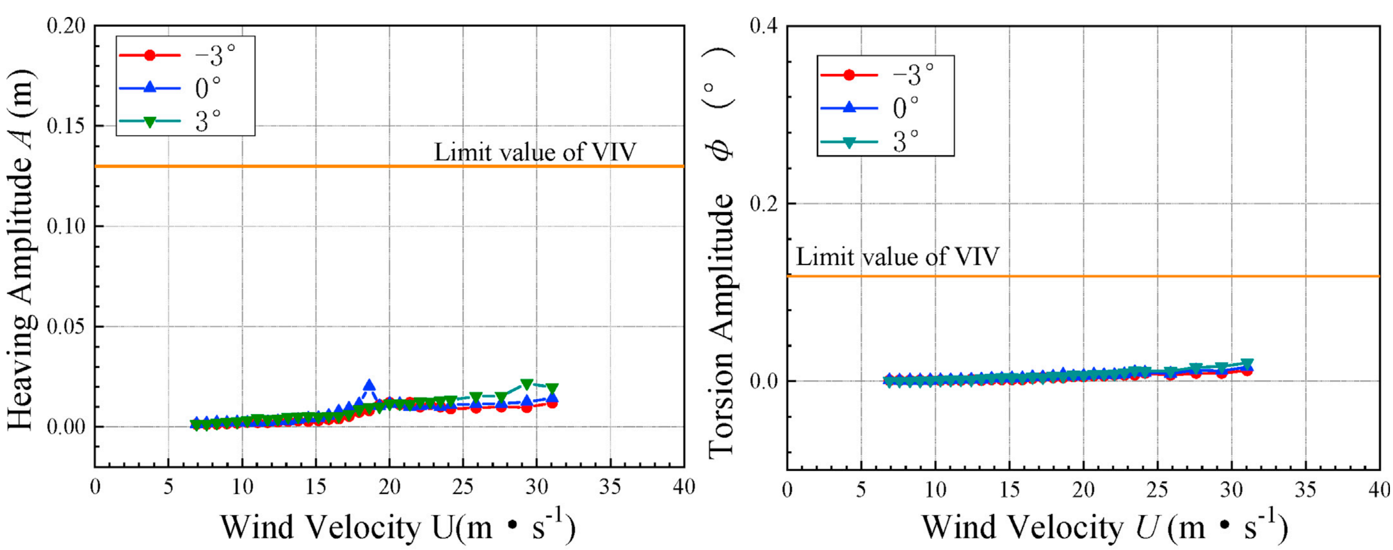
Table 17 shows the flutter wind tunnel test results of Liujiaxia Bridge with irregularly sealed side traffic barrier measures. For comparison purpose, the results associated with Case 10 are also presented here. By comparing Cases 10 and 11, it is noted that: (1) Although the total height for the barrier is the same, they mainly differ in the sealing strategy and ventilation rate, 35% and 60% for Cases 10 and 11, respectively. Increasing the ventilation rate of the traffic barrier has a negligible effect on the flutter critical wind speed at wind attack angles -3°, 0° and 3°, but it leads to a larger flutter critical wind speed at the positive wind attack angle 5° and a smaller one at the negative wind attack angle -5°. Therefore, the efficacy of modifying the ventilation rate to improve the flutter stability would vary with the wind attack angle and sectional

(c) E-①　　(d) E-③

Fig. 35. Side traffic barrier and sidewalk traffic barrier for the section model of Chongqing Hechuan Bridge.

Table 16
VIV and flutter wind tunnel test results associated with the sealed central traffic barrier, sealed side traffic barrier, and sealed sidewalk traffic barrier measures.

Case	Aerodynamic measure	wind attack angle					
		+3°		0°		-3°	
		VIV	Flutter	VIV	Flutter	VIV	Flutter
IV	C-③	observed	44.5	–	–	–	–
V	A+ (C-③)	observed	50.3	–	–	–	–
VI	A+(B-①)	observed	40.2	–	–	–	–
VII	(D-②)+(E–②)	observed	51.7	–	–	–	–
VIII	A+(D-②)+(E–②)	observed	55.8	–	–	–	–
IX	A+(D-②)+(E–②)+(C-③)+(B-①)	No VIV	>63.5	No VIV	>63.5	No VIV	>63.5
X	(E–③)+(D-①) +(B-②) + (C-②)	No VIV	56.3	No VIV	>63.5	No VIV	>63.5



(a) Vertical response　　(b) Torsional response

Fig. 36. Vertical and torsional VIV responses of Chongqing Hechuan Bridge under the measures in Case Ⅸ

aerodynamic shape. As for the Liujiaxia Bridge, the flutter critical wind speeds associated with Case 10, with the ventilation rate 35%, under various wind attack angles satisfy the code requirements. This illustrates that for different section models, there would exist a satisfactory ventilation rate such that the flutter stability at various wind attack angles can be well optimized, and this ventilation rate can be obtained via wind

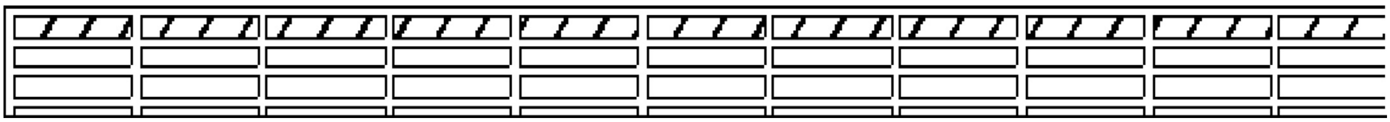


Fig. 37. Horizontally sealed traffic barrier measure for the Hongkong-Zhuhai-Macao Bridge (Measure 3, following the naming in Fig. 11).

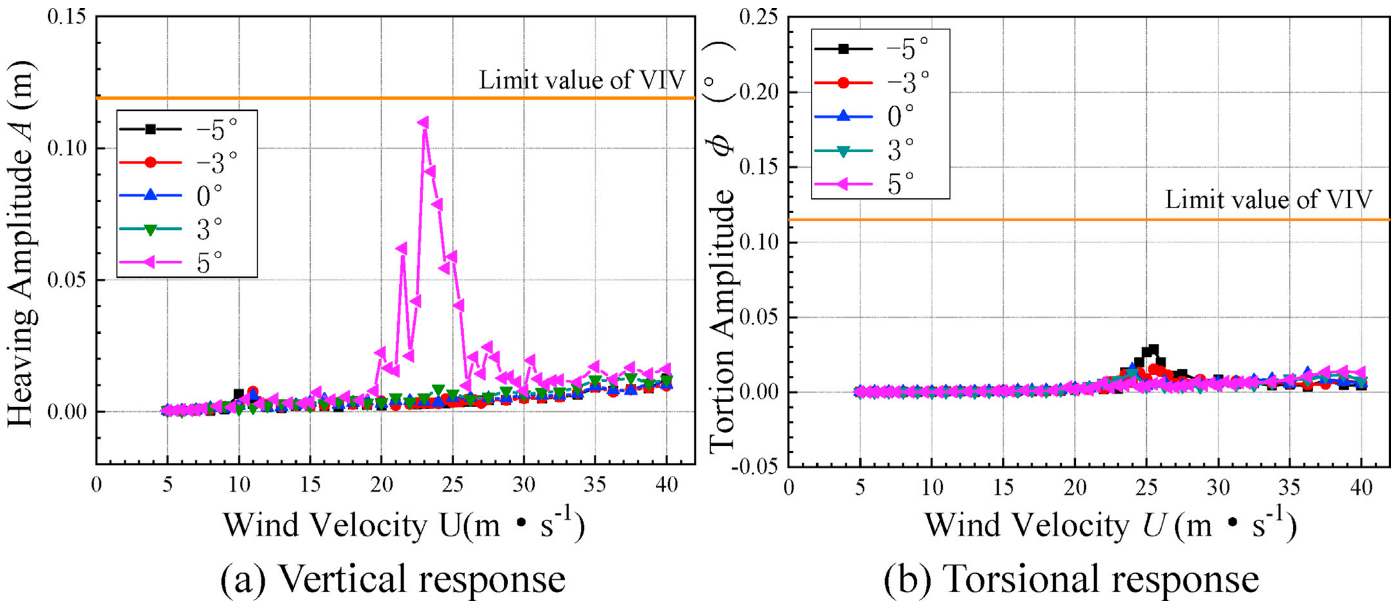


Fig. 38. Vertical and torsional VIV responses for Measure 3.

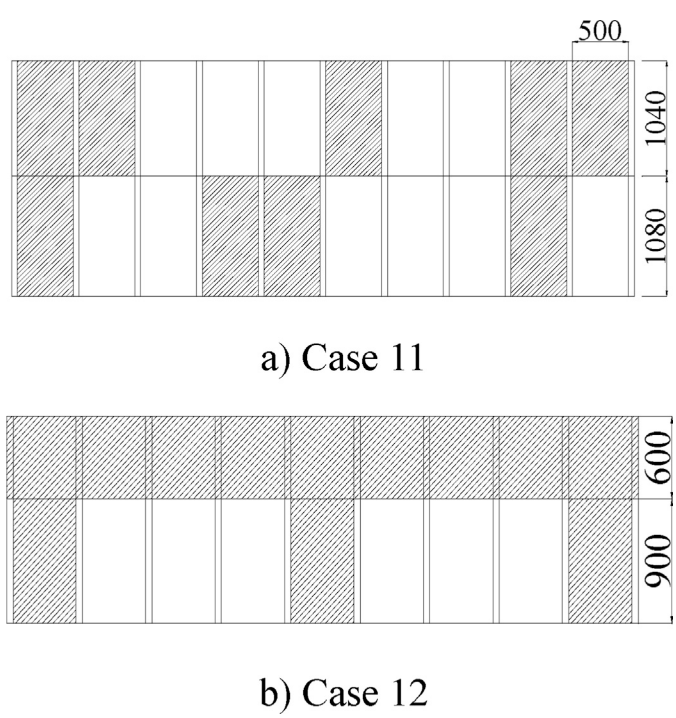


Fig. 39. Sealed side traffic barrier measures for the Liujiaxia Bridge (Unit: mm).

tunnel tests; and (2) Case 10 results in a more balanced flutter stability at different wind attack angles than Case 11. As such, the efficacy in improving the flutter performance at each wind attack angle associated with the irregularly sealed traffic barrier is lower than that associated with the regular sealed traffic barrier.

Table 17
Flutter wind tunnel test results of Liujiaxia Bridge with irregularly sealed side traffic barrier measures.

Case	Flutter critical wind speed (m/s)				
	-5°	-3°	0°	3°	5°
10	66.0	>70.4	66.0	>70.4	61.6
11	57.2	>70.4	66.0	>70.4	>70.4
12	>57.2	>57.2	>57.2	>57.2	50.6

Note: For comparison purpose, the results associated with Case 10 are also presented here.

In comparisons between Cases 10 and 12, with the ventilation rate being 35% and 45%, respectively, it can be seen that, in Case 12, the flutter stability at different wind attack angles is slightly decreased. This is due to that although the sealing form of the traffic barrier being the same, the height of the traffic barrier is reduced in Case 12. Therefore, a higher height traffic barrier should be advocated if conditions permit.

6.3. Effect of arc-shaped and regular seals on flutter and VIV

In this section, expanded research is conducted regarding the aerodynamic measures for the Chongqing Hechuan Bridge. As discussed in the previous section, the combined measures in Case IX can effectively improve the VIV and flutter stability performance of the original scheme of the Chongqing Hechuan Bridge. Considering the landscape effect of the bridge deck, efforts has been put to alter the way of sealing the side traffic barrier and sidewalk traffic barrier without changing the ventilation rate. For this purpose, the arc-shaped sealing form is optimized to attain a better aesthetic view. Fig. 40 shows the schematic diagram of the side traffic barrier (D-③) and sidewalk traffic barrier (E−④) correspondingly optimized for measures D-② and E−② and Fig. 41 shows the corresponding test photographs.

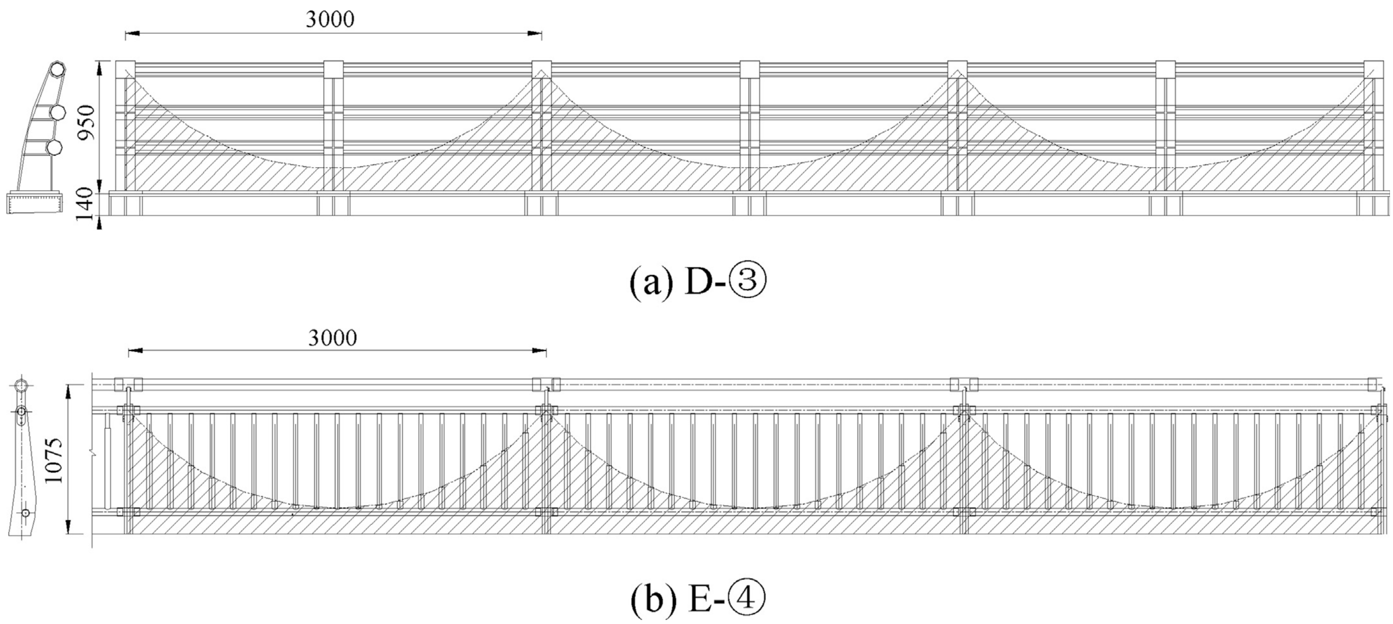


Fig. 40. Schematic diagram of arc-shaped sealing for the side traffic barrier and sidewalk traffic barrier optimization (Unit: mm).

Table 18 shows the wind tunnel test results for two more cases with the optimized side traffic barrier and sidewalk traffic barrier, where the bridge VIV can be observed in these two additional cases. The results show that although the ventilation rate of the side traffic barrier and sidewalk traffic barrier remains unchanged, the side traffic barrier (D-③) and the sidewalk traffic barrier (E−④) in an arc-shaped sealed form cannot achieve a favorable vibration suppression effect. As such, the sealing form of the side traffic barrier and sidewalk traffic barrier has a substantial influence on the VIV and flutter stability of the bridge. The form of the sealed traffic barrier with regular vertical intervals performs better for the vibration suppression effect of VIV than the arc-shaped sealed form.

7. Concluding remarks

In the present paper, an alternative aerodynamic mitigation measure, i.e., sealed traffic barrier, for bridges was proposed and three distinctive long-span bridges have been selected as research projects to investigate the efficacy of this measure in improving the flutter and vortex induced vibration (VIV) stability of bridges through wind tunnel tests. The performance of the sealed traffic barrier with various sealing forms on the bridge aerodynamic stability is investigated and summarized. Theoretically speaking, partially sealed bridge deck traffic barrier or sidewalk traffic barrier would interfere with the occurrence of periodic vortex shedding on the bridge surface, suppress the VIV, and improve the flutter stability of the bridge. In addition, when common mitigation measures such as the fairing, edge plate, flaps, etc. cannot improve the wind-induced vibration of bridge, the measures of fully or partially sealed traffic barrier or sidewalk traffic barrier can be installed, without causing significant additional structural costs. This proposed measure can potentially contribute to the wind-resistance design of long-span bridges. The following conclusions can be drawn:

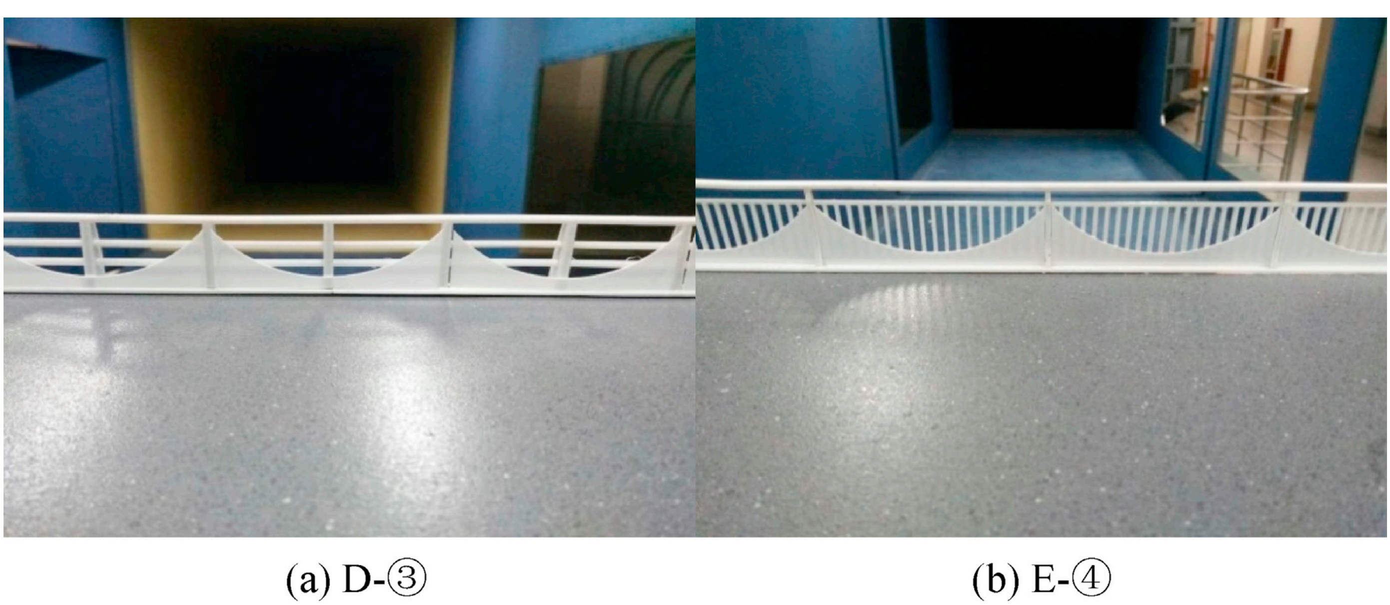


Fig. 41. Photos of the arc-shaped sealing for the side traffic barrier and sidewalk traffic barrier.

Table 18
VIV and flutter wind tunnel test results for the optimized side traffic barrier and sidewalk traffic barrier.

Case	Aerodynamic measure	wind attack angle					
		+3°		0°		-3°	
		VIV	Flutter	VIV	Flutter	VIV	Flutter
XI	(E−③)+ (D-③)+(B-②) + (C-①)	observed	42.6	–	–	–	–
XII	(E−④)+ (D-③)+(B-②)+ (C-③)	observed	48.2	–	–	–	–

(1) The results show that the measures associated with the traffic barrier along the bridge deck sealed with regular vertical intervals have significant influence on the flutter and VIV performance of the bridge. Through reasonable design of the sealing form for the traffic barrier, both the bridge flutter and VIV stability can be improved simultaneously.
(2) The sealed traffic barrier or sidewalk traffic barrier can play similar role to the upward central stabilizer in improving the bridge flutter and VIV performance. For the steel truss bridge, the flutter vibration suppression effect of the guide plate is not as good as that of the central stabilizer. When the bridge deck is not suitable for setting the upward central stabilizer, the measures of heightening and sealing the side traffic barrier can result in similar effects to the upward central stabilizer.
(3) Regarding the effects of the sealing form for the traffic barrier on the bridge VIV and flutter performance, the flutter stability at different wind attack angles can slightly decrease when the height of the traffic barrier is reduced, though the sealing form of the traffic barrier is the same. Increasing the ventilation rate of the traffic barrier has a negligible effect on the flutter critical wind speed at small wind attack angles, but it leads to a larger flutter critical wind speed at large positive wind attack angles and a smaller one at large negative wind attack angles. The form of the sealed traffic barrier with regular vertical intervals can perform better for the vibration suppression effect of VIV than the arc-shaped sealed form.

The limitations of the present study and future work are given as follows:

(1) This study mainly justified the proposed sealed traffic barrier measure and verified its efficacy on three typical wind-sensitive long-span bridges. However, more studies should be conducted to further elaborate its working mechanism on suppressing the bridge vibration.
(2) Other than the ventilation rate, the sealing form is another key factor for optimizing the aerodynamic shape of the bridge girder. Both the ventilation rate and the sealing form should be considered and optimized for achieving appreciable suppression effect.

CRediT authorship contribution statement

Hua Bai: Conceptualization, Methodology, Writing - original draft, Visualization, Formal analysis, Investigation, Project administration, Funding acquisition. **Naichuan Ji:** Investigation, Software, Validation, Formal analysis. **Guoji Xu:** Visualization, Data curation, Writing - review & editing. **Jiawu Li:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

Bruno, L., Mancini, G., 2002. Importance of deck details in bridge aerodynamics. Struct. Eng. Int. 12, 289–294. https://doi.org/10.2749/101686602777965234.
Chen, X.Z., Kareem, A., 2006. Revisiting multi-mode coupled bridge flutter: some new insights. J. Eng. Mech. 132 (10), 1115–1123. https://doi.org/10.1061/(ASCE)0733-9399.
Chen, A.R., Zhou, Z.Y., Xiang, H.F., 2006. On the mechanism of vertical stabilizer plates for improving aerodynamic stability of bridges. Wind Struct. 9 (1), 59–74. https://doi.org/10.12989/was.2006.9.1.059.
Diana, G., Fiammenghi, G., Belloli, M., Rocchi, D., 2013. Wind tunnel tests and numerical approach for long span bridges: the Messina Bridge. J. Wind Eng. Ind. Aerod. 122, 38–49. https://doi.org/10.1016/j.jweia.2013.07.012.
Fujino, Y., Siringoringo, D., 2013. Vibration mechanisms and controls of long-span briges: a review. Struct. Eng. Int. 23 (3), 248–268. https://doi.org/10.2749/101686613X13439149156886.
Fujino, Y., Yoshida, Y., 2002. Wind-induced vibration and control of Trans-Tokyo Bay crossing bridge. J. Struct. Eng. 128 (8), 1012–1025. https://doi.org/10.1061/(ASCE)0733-9445.
Ge, Y.J., Xiang, H.F., 2008. Bluff body aerodynamics application in challenging bridge span length. In: Bluff Bodies Aerodynamics & Applications (BBAA-VI) (Milana, Italy).
Ge, Y.J., Zou, X.J., Yang, Y.X., 2009. Aerodynamic stabilization of central stabilizers for box girder suspension bridges. Wind Struct. 12 (4), 285–298. https://doi.org/10.12989/was.2009.12.4.285.
Irwin, P.A., 2008. Bluff body aerodynamics in wind engineering. J. Wind Eng. Ind. Aerod. 96 (6,7), 701–712. https://doi.org/10.1016/j.jweia.2007.06.008.
Jones, N.P., Scanlan, R.H., Sarkar, P.P., Singha, L., 1995. The effect of section model details on aeroelastic parameters. J. Wind Eng. Ind. Aerod. 54–55, 45–53. https://doi.org/10.1016/0167-6105(94)00028-C.
Kubo, Y., Sadashima, K., Yamaguchi, E., Kato, K., Okamoto, Y., Koga, T., 2001. Improvement of aeroelastic instability of shallow π section. J. Wind Eng. Ind. Aerod. 89 (14–15), 1445–1457. https://doi.org/10.1016/S0167-6105(01)00151-9.
Kubo, Y., Kimura, K., Sadashima, K., Sadashima, K., Okamoto, Y., Yamaguchi, E., Kato, K., 2002. Aerodynamic performance of improved shallow π shape bridge deck. J. Wind Eng. Ind. Aerod. 90 (12–15), 2113–2125. https://doi.org/10.1016/S0167-6105(02)00328-8.
Larsen, A., 1993. Aerodynamic aspects of the final design of the 1624 m suspension bridge across the Great Belt. J. Wind Eng. Ind. Aerod. 48, 261–285. https://doi.org/10.1016/0167-6105(93)90141-A.
Larsen, A., 1998. Advances in aeroelastic analysis of suspension and cable-stayed bridges. J. Wind Eng. Ind. Aerod. 74–76, 73–90. https://doi.org/10.1016/S0167-6105(98)00007-5.
Larsen, A., 2008. Aerodynamic stability and vortex shedding excitation of suspension bridges. In: Proceedings of the 4th International Conference on Advances in Wind and Structures (AWAS'08). Techno Press, Jeju, Korea.
Larsen, A., Larose, G.L., 2015. Dynamic wind effects on suspension and cable-stayed bridges. J. Sound Vib. 334, 2–28. https://doi.org/10.1016/j.jsv.2014.06.009.
Larsen, A., Wall, A., 2012. Shaping of bridge box girders to avoid vortex shedding response. J. Wind Eng. Ind. Aerod. 104–106, 159–165. https://doi.org/10.1016/j.jweia.2012.04.018.
Larsen, A., Esdahl, S., Andersen, J.E., Vejrum, T., 2000. Storebælt suspension bridge-vortex shedding excitation and mitigation by guide vanes. J. Wind Eng. Ind. Aerod. 88 (2), 283–296. https://doi.org/10.1016/S0167-6105(00)00054-4.
Li, M., Sun, Y.G., Jing, H.M., Mingshui, L., 2018. Vortex-induced vibration optimization of a wide streamline box girder by wind tunnel test. J. Civ. Eng. 22 (12), 5143–5153. https://doi.org/10.1007/s12205-018-0548-y.
Matsumoto, M., Yoshizumi, F., Yabutani, T., Yabutani, Y., Abe, K., Nakajima, N., 1999. Flutter stabilization and heaving-branch flutter. J. Wind Eng. Ind. Aerod. 83 (1–3), 289–299. https://doi.org/10.1016/S0167-6105(99)00079-3.

Ministry of Communications of the People's Republic of China, 2004. JTG/TD60-01-2004 Wind-Resistant Design Specification for Highway Bridges. China Communications Press, Beijing, China.
Miyata, T., 2003. Historical view of long-span bridge aerodynamics. J. Wind Eng. Ind. Aerod. 91, 1393–1410. https://doi.org/10.1016/j.jweia.2003.09.033.
Miyata, T., Yamaguchi, K., 1993. Aerodynamics of wind effects on the Akashi Kaikyo bridge. J. Wind Eng. Ind. Aerod. 48 (2–3), 287–315. https://doi.org/10.1016/0167-6105(93)90142-B.
Nagao, F., Utsunomiya, H., Kawase, A., Yoshioka, E., 1995. Effects of handrails on vortex induced oscillation of box girder bridge. In: Proceedings of the 6th International Conference on Flow-Induced Vibration (London, UK).
Nagao, F., Utsunomiya, H., Yoshioka, E., Ikeuchi, A., Kobayashi, H., 1997. Effects of handrails on separated shear flow and vortex-induced oscillation. J. Wind Eng. Ind. Aerod. 69–71, 819–827. https://doi.org/10.1016/S0167-6105(97) 00208-0.
Saito, T., Sakata, H., 1999. Aerodynamic stability of long-span box girder bridges and anti-vibration design considerations. J. Fluid Struct. 13, 999–1016. https://doi.org/10.1006/jfls.1999.0238.
Sakai, Y., Ogawa, K., Shimodoi, H., Saitoh, T., 1993. An experimental study on aerodynamic improvements for edge girder bridges. J. Wind Eng. Ind. Aerod. 49 (1–3), 459–466. https://doi.org/10.1016/0167-6105(93)90040-U.
Sato, H., Kusuhara, S., Ogi, K., Matsufuji, H., 2000. Aerodynamic characteristics of super long-span bridges with slotted box girder. J. Wind Eng. Ind. Aerod. 88, 297–306. https://doi.org/10.1016/S0167-6105(00)00055-6.
Tang, H.J., Li, Y.L., Shum, K.M., 2018. Flutter performance and aerodynamic mechanism of plate with central stabilizer at large angles of attack. Adv. Struct. Eng. 21 (3), 335–346. https://doi.org/10.1177/1369433217717120.
Vaz, D.C., Almeida, R.A.B., Didier, E., Urgueira, A.P.V., Borges, A.R.J., 2016. Improving the aerodynamic performance of Vila-Real Bridge deck-section. J. Wind Eng. Ind. Aerod. 156, 72–83. https://doi.org/10.1016/j.jweia.2016.07.002.
Wang, Q., Liao, H., Li, M., Xian, R., 2009. Wind tunnel study on aerodynamic optimization of suspension bridge deck based on flutter stability. In: 7th Asia-Pacific Conference on Wind Engineering (APCWE-VII). Taipei, Taiwan.
Weight, A.J., 2009. Critical analysis of the great belt east bridge Denmark. In: Proceedings of Bridge Engineering 2 Conference 2009 (Bath, UK).
Wu, T., Kareem, A., 2012. An overview of vortex-induced vibration (VIV) of bridge decks. Front. Struct. Civ. Eng. 6, 335–347. https://doi.org/10.1007/s11709-012-0179-1.
Yang, Y.X., Ge, Y.J., 2009. Aerodynamic flutter control for typical girder sections of long-span cable-supported bridges. Wind Struct. 12 (3), 205–217. https://doi.org/10.12989/was.2009.12.3.205.
Yang, Y.X., Wu, T., Ge, Y.J., Kareem, A., 2015. Aerodynamic stabilization mechanism of a twin box girder with various slot widths. J. Bridge Eng. 20 (3), 04014067 https://doi.org/10.1061/(ASCE)BE.1943-5592.0000645.
Yang, Y.X., Zhou, R., Ge, Y., Zou, X., Zhang, L.H., 2017. Flutter characteristics of twin-box girder bridges with vertical central stabilizers. Eng. Struct. 133, 33–48. https://doi.org/10.1016/j.engstruct.2016.12.009.
Yang, Y.X., Zhang, L., Ding, Q.S., Ge, Y.J., 2018. Flutter performance an improvement for a suspension bridge with central-slotted box girder during erection. J. Wind Eng. Ind. Aerod. 179, 118–124. https://doi.org/10.1016/j.jweia.2018.05.016.
Zhou, R., Yang, Y., Ge, Y.J., Zhang, L., 2018. Comprehensive evaluation of aerodynamic performance of twin-bow girder bridges with vertical stabilizers. J. Wind Eng. Ind. Aerod. 175, 317–327. https://doi.org/10.1016/j.jweia.2018.01.039.
Zhu, L.D., Tan, X., Guo, Z.S., Ding, Q.S., 2019. Effects of central stabilizing barrier on flutter performances of a suspension bridge with a truss-stiffened deck under skew winds. Adv. Struct. Eng. 22 (1), 17–29. https://doi.org/10.1177/1369433218774144.